9 Laplace Transforms

Objective

After completing this chapter, user will be able to understand:

- Concept of ROC and its importance in Laplace transform
- Bilateral Laplace transform of standard signal
- Properties of ROC and BLT
- Inverse Laplace transform
- Unilateral Laplace transform and its properties
- Solution of differential equations
- Block diagram representation and system realisation

Introduction

- The Laplace transform possesses a distinct set of properties parallel to those of the Fourier transform.
- The Laplace transform can be applied to the analysis of many unstable systems and consequently plays an important role in the investigation of the stability or instability of systems.
- The Laplace transform comes in two varieties. Bilateral, or two-sided Laplace transform, and unilateral, or one-sided Laplace transform.
- The bilateral Laplace transform offers insight into the nature of system characteristics such as stability, causality, and frequency response.
- The unilateral Laplace transform is a convenient tool for solving differential equations with initial conditions.

Laplace Transform

Prerequisite:

Right sided signal (RSS.):

1. Duration of signal → Infinite
2. x (t) must be zero toward left of some $t = t_0$

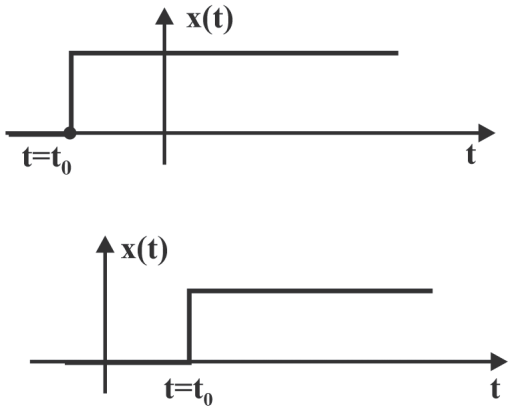


Fig. 9.1

Left Sided Signal (LSS):

1. Duration of signal → Infinite.
2. x (t) must be zero towards right of some $t = t_0$.

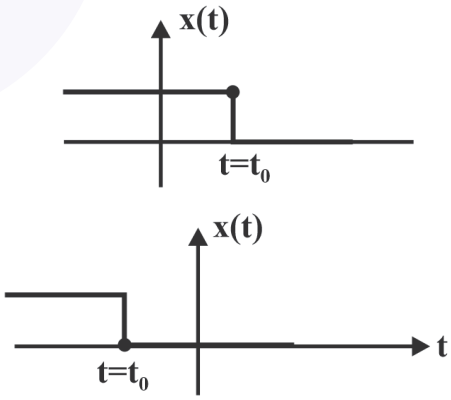


Fig. 9.2

Both sided or double sided signals:

1. Duration of signal → Infinite
2. x(t) → around $t = t_0$: LSS + RSS

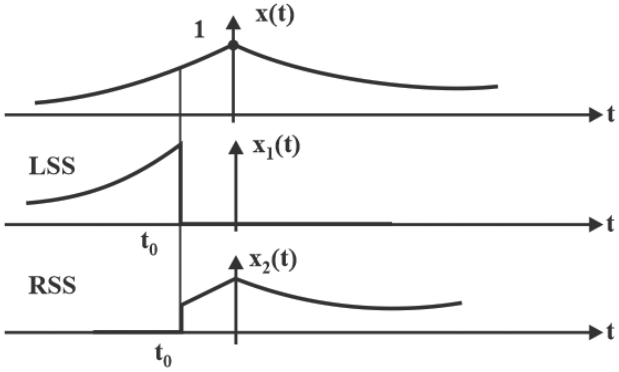


Fig. 9.3

x (t) is the finite duration and absolutely integrable signal:

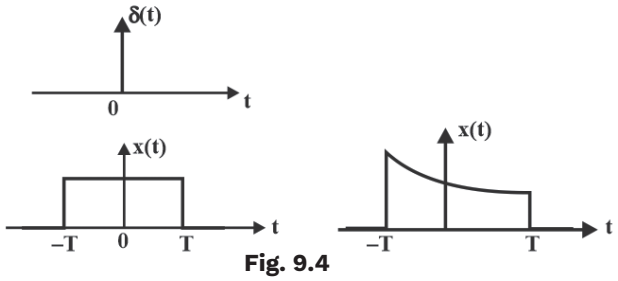


Fig. 9.4

Rectangular complex plane: S plane:

1. $S = \sigma + j\omega$ $\quad : -\infty < \sigma < \infty$
   $-\infty < \omega < \infty$
2. $\text{Re}\{s\} = \sigma$
3. $\text{Im}\{s\} = \omega$

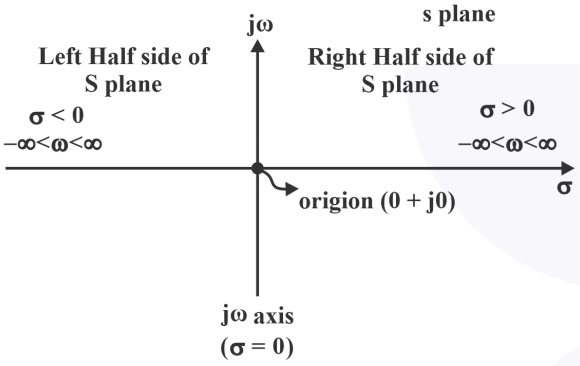


Fig. 9.5

$\sigma = 0$: $j\omega$ *axis*

$\sigma > 0$: *R.H.*S of s plane

$\sigma < 0$: *L.H.*S of s plane

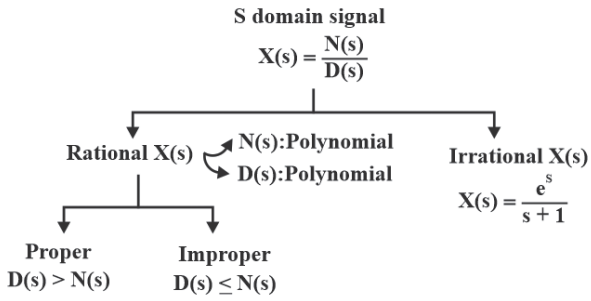


Fig. 9.6

Poles and zeros of s domain signal:

$$S = S_z \text{ is Zero of } X(S) = \frac{N(S)}{D(S)} \to \lim_{S \to S_z} X(S) = 0$$

$$S = S_p \text{ is Pole of } X(S) =$$

$$\frac{N(S)}{D(S)} \to \lim_{S \to S_p} X(S) = \pm\infty$$

1. $X(S) = \dfrac{e^S - e^{-S}}{S}$

Is $S = 0$ *pole of* $X(S) = ?$

Solution:

$$\lim_{S \to 0} X(S) = \lim_{S \to 0} \left\{ \frac{e^S - e^{-S}}{S} \right\} = \frac{0}{0}$$

$$= \lim_{S \to 0} \frac{\left[e^S + e^{-S} \right]}{1} = 2$$

$S = 0$ *is not pole of* $X(S)$.

$$X(S) = \frac{\left\{ 1 + S + \frac{S^2}{2!} + \cdots \right\} + \left\{ -1 + S - \frac{S^2}{2!} + \cdots \right\}}{S}$$

$$X(S) = 2 + \frac{2S^2}{3!} + \ldots\ldots \to \text{Polynomial}.$$

2. $X(S) = \dfrac{(S+1)^2}{(S+1)^2 (S-2)(S+3)}$

Solution:

S = -1 → Nota pole

S = 2 → pole

S = -3 → pole

Note: Pole and zeros will be calculated after common factor cancellation.

Pole Mapping on S Plane from X (S)

$$X(S) = \frac{1}{S+1}$$

Pole: S = -1 → Point

Re {S} = -1 = σ → Line

Re {S} > -1 → 2-D Region.

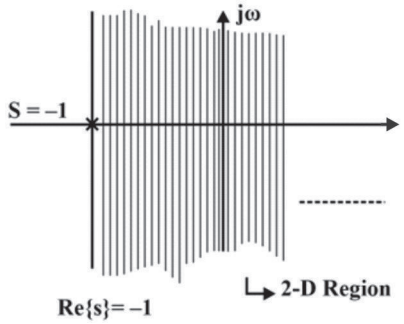


Fig. 9.7

3. $X(S) = \dfrac{1}{S^2 + S + 1}$

Solution:

$$X(S) = \frac{1}{S^2 + S + 1}$$

$$S = -\frac{1}{2} + j\frac{\sqrt{3}}{2} \rightarrow Point$$

$$\text{Re}\{S\} = -\frac{1}{2} \rightarrow Line$$

$$\text{Re}\{S\} > -\frac{1}{2} \rightarrow 2 - D \text{ Re}gion.$$

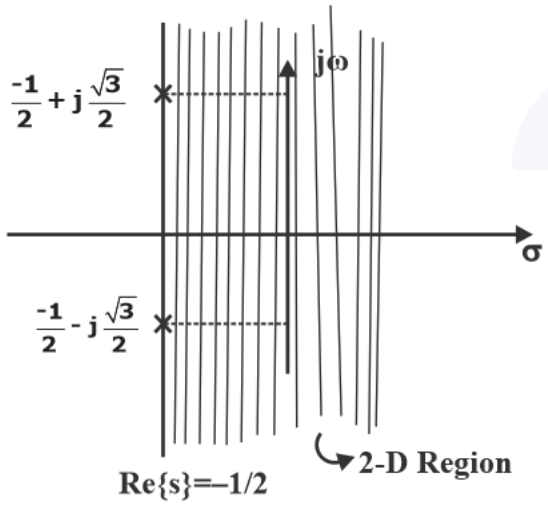


Fig. 9.8

Writing X(S) from the pole-zero diagram.

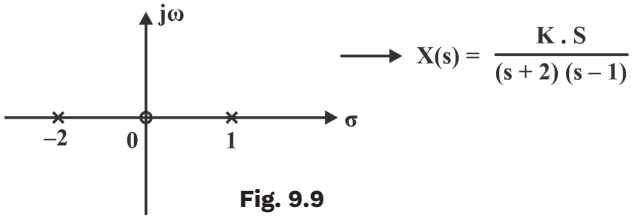


Fig. 9.9

Figure: Pole zero diagram:

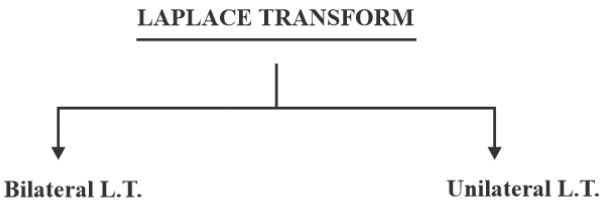


Fig. 9.10

$$X(S) = \int_{-\infty}^{\infty} x(t)e^{-St}dt \text{ --- Bilateral L.T.}$$

$$X(S) = \int_{0^-}^{\infty} x(t)e^{-St}dt \text{ --- Unilateral L.T}$$

Pair: $x(t) \xleftrightarrow{B.L.T.} X(S)$

Formula: $X(S) = \int_{-\infty}^{\infty} x(t)e^{-St}dt$

Relation with F.T.

$$X(S) = \int_{-\infty}^{\infty} \left(x(t)e^{-\sigma t}\right)e^{-j\omega t}dt = F.T.\left[\underbrace{x(t)e^{-\sigma t}}\right]$$

Bilateral Laplace transform involves an integration from -∞ to +∞, while the unilateral Laplace transform has limits of integration from 0^- to +∞.

Procedure to Calculate BLT

S.1 $x(t)e^{-\sigma t}$ must be absolutely integrable, for BLT to converge.

S.2 If BLT converges: $X(S) = \int_{-\infty}^{\infty} x(t)e^{-St}dt$

4. Calculate BLT of

$$x(t) = e^{-S_0 t}u(t) \quad : \quad S_0 = \sigma_0 + j\omega.$$

Solution:

S.1

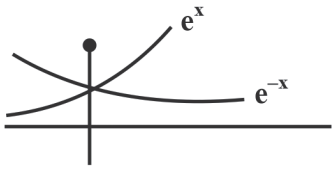
$$\int_{-\infty}^{\infty} \left|x(t)e^{-\sigma t}\right|dt \rightarrow ? \qquad |z_1 z_2| = |z_1||z_2|$$

$$\int_{-\infty}^{\infty} \left|e^{-S_0 t}u(t)e^{-\sigma t}\right|dt = \int_{-\infty}^{\infty} \left|e^{-(\sigma+\sigma_0)t}e^{-j\omega t}u(t)\right|dt$$

$$= \int_{-\infty}^{\infty} \left|e^{-(\sigma+\sigma_0)t}\right| \left|e^{-j\omega t}\right| \left|u(t)\right| dt$$

$$= \int_{0}^{\infty} \left|e^{-(\sigma+\sigma_0)t}\right| dt$$

$$= \int_{0}^{\infty} e^{-(\sigma+\sigma_0)t} dt$$



$$= \left\{ \frac{e^{-(\sigma+\sigma_0)t}}{-(\sigma+\sigma_0)} \right\}_0^{\infty} = \left\{ \frac{1 - e^{-(\sigma+\sigma_0)\infty}}{\sigma+\sigma_0} \right\}$$

$\sigma+\sigma_0$	$x(t)e^{-\sigma t}$	$X(S)$
$\sigma+\sigma_0 > 0$	*Abs* $\int$	*Formula*
$\sigma+\sigma_0 = 0$	*Abs Not* $\int$	$X(s)$ does not converge
$\sigma+\sigma_0 < 0$	*Abs Not* $\int$	

S.2 If : $\sigma+\sigma_0 > 0$

$\text{Re}\{S\} > -\sigma_0 \rightarrow 2-D$ Re*gion*

Then X(S) converge

$$X(S) = \int_{-\infty}^{\infty} x(t)e^{-St}dt = \int_{0}^{\infty} e^{-(S+S_0)t}dt = \frac{1}{S+S_0}$$

$$e^{-S_0 t}u(t) \xleftrightarrow{B.L.T} \begin{cases} X(S) = \dfrac{1}{S+S_0} : only\ if\ \overbrace{\text{Re}\{S\} > -\sigma_0}^{2-D\ \text{Re}gion} \\ \infty \quad : elsewhere\ \ \text{Re}\{S\} \le -\sigma_0 \end{cases}$$

Conclusion:

$$x(t) = e^{-S_0 t}u(t) \longleftrightarrow X(S) = \frac{1}{S+S_0}$$

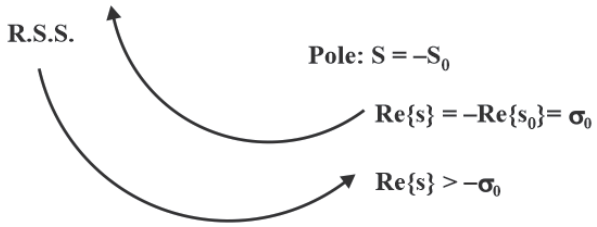
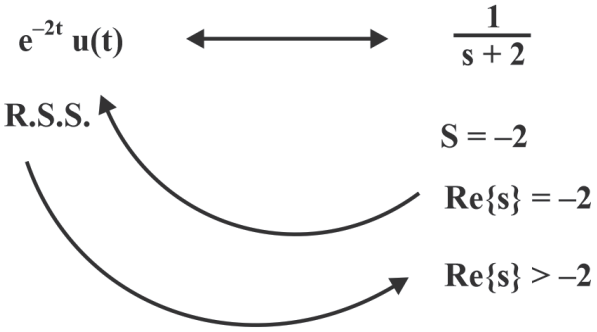


Fig. 9.11



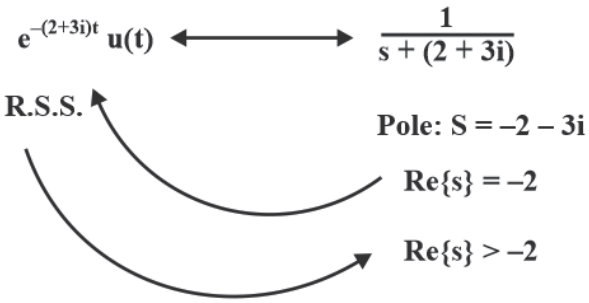


Fig. 9.12

$$e^{(3-3i)t}u(t) \longleftrightarrow \frac{1}{S+(-3+3i)}$$

Pole: $S = 3-3i$

$\text{Re}\{S\} = 3 : R \propto : \text{Re}\{S\} > 3$

Points to Remember

$x(t)$	$X(S)$	*ROC*
$e^{-s_0 t}u(t)$	$\dfrac{1}{S+S_0}$	$\text{Re}\{S\} > -\text{Re}\{S_0\}$
$e^{s_0 t}u(t)$	$\dfrac{1}{S-S_0}$	$\text{Re}\{S\} > \text{Re}\{S_0\}$
$e^{-at}u(t)$	$\dfrac{1}{S+a}$	$\text{Re}\{S\} > -a$
$e^{at}u(t)$	$\dfrac{1}{S-a}$	$\text{Re}\{S\} > a$
$u(t)$	$\dfrac{1}{S}$	$\text{Re}\{S\} > 0$

Where a is real.

5. Calculate BLT of $x(t) = -e^{-S_0 t}u(-t)$

Solution:

S.1 $\int_{-\infty}^{\infty} \left|x(t)e^{-\sigma t}\right| dt = \int_{-\infty}^{\infty} \left|-e^{-\sigma_0 t}e^{-j\omega_0 t}u(-t)e^{-\sigma t}\right| dt$

$= \int_{-\infty}^{\infty} \left|e^{-(\sigma+\sigma_0)t}\right| \left|e^{-j\omega_0 t}\right| \left|u(-t)\right| dt$

$= \int_{-\infty}^{0} e^{\underbrace{-(\sigma+\sigma_0)t}} dt = \left\{\frac{e^{\overbrace{+(\sigma+\sigma_0)(+\infty)}^{\sigma+\sigma_0<0}} - 1}{\sigma+\sigma_0}\right\}$

S.2 If $\sigma + \sigma_0 < 0$

$\text{Re}\{S\} < -\sigma_0$: $2-D$ Region

Then X(S) converges:

$$X(S) = \int_{-\infty}^{\infty} -e^{-S_0 t}u(-t)e^{-St}dt = \frac{1}{S+S_0}$$

$$e^{-S_0 t}u(t) \longleftrightarrow X(S) = \begin{cases} \frac{1}{S+S_0} : \text{Re}\{S\} > \text{Re}\{-S\} \\ \infty \quad : else \end{cases}$$

$$-e^{-S_0 t}u(-t) \longleftrightarrow X(S) = \begin{cases} \frac{1}{S+S_0} : \text{Re}\{S\} > \text{Re}\{-S\} \\ \infty \quad : else \end{cases}$$

Points to Remember

$x(t)$	$X(S)$	$ROC:$
$-e^{-s_0 t}u(-t)$	$\frac{1}{S+S_0}$	$\text{Re}\{S\} < -\text{Re}\{S_0\}$
$-e^{s_0 t}u(-t)$	$\frac{1}{S-S_0}$	$\text{Re}\{S\} < \text{Re}\{S_0\}$
$-e^{-at}u(-t)$	$\frac{1}{S+a}$	$\text{Re}\{S\} < -a$
$-e^{at}u(-t)$	$\frac{1}{S-a}$	$\text{Re}\{S\} < a$
$-u(-t)$	$\frac{1}{S}$	$\text{Re}\{S\} < 0$

Where a is real.

Properties of BLT

Linearity: $x_1(t) \longleftrightarrow X_1(S)$ $\quad ROC : R_1$

$x_2(t) \longleftrightarrow X_2(S)$ $\quad ROC : R_2$

Case 1:

$$\underbrace{x(t) = Ax_1(t) + Bx_2(t)}_{\substack{R.S.S. \\ L.S.S. \\ B.S.S.}} \longleftrightarrow R_1 \cap R_2 =$$

$\{\phi\} : X(S) : D.N.E.$

$R_1 \cap R_2 \neq \{\phi\} : X(S) = AX_2(S) + BX_2(S)$

$ROC : R_1 \cap R_2$

Case 2:

$$\underbrace{x(t) = Ax_1(t) + Bx_2(t)}_{F.D.+Abs\int} \longleftrightarrow X(S) =$$

$AX_1(S) + BX_2(S)$

ROC : Entire S Plane.

Note: If a signal is F.D. + Abs ∫ then ROC will be entire S plane.

6. Calculate BLT of $x(t) = u(t) - u(t-T)$.

Solution:

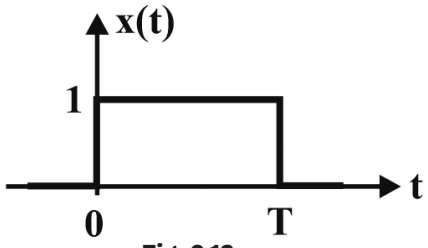


Fig. 9.13

S.1 $\int_{-\infty}^{\infty} \left|x(t)e^{-\sigma t}\right| dt = \int_{0}^{T} \left|1.e^{-\sigma t}\right| dt = \frac{\left\{1-e^{-\sigma T}\right\}}{\sigma}$

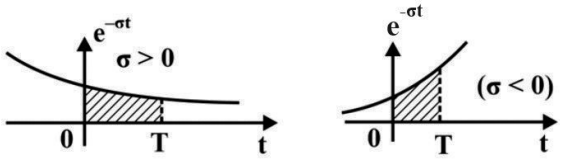


Fig. 9.14

7. $x(t) = e^{-2|t|}$. Calculate BLT

Solution:

$$x(t) = \underbrace{e^{2t}u(-t)}_{LSS} + \underbrace{e^{-2t}u(t)}_{RSS}$$

$$x_1(t) = e^{2t}u(-t) \longleftrightarrow X_1(S) = \frac{-1}{S-2} \quad R_1 : \text{Re}\{S\} < 2$$

$$x_2(t) = e^{-2t}u(t) \longleftrightarrow X_2(S) = \frac{1}{S+2} \quad R_2 : \text{Re}\{S\} > -2$$

$$x(t) = \underbrace{x_1(t) + x_2(t)}_{B.S.S} \longleftrightarrow R_1 \cap R_2 \neq \{\phi\} : X(S) = \frac{-1}{S-2} + \frac{1}{S+2}$$

$$ROC : -2 < \text{Re}\{S\} < 2$$

8. $x(t) = e^{2|t|}$. Calculate BLT

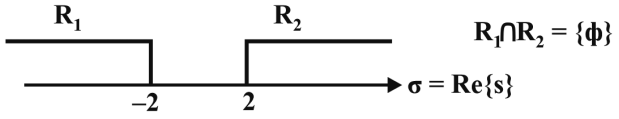
Solution:

$$x(t) = \underbrace{e^{-2t}u(-t)}_{x_1(t)} + \underbrace{e^{2t}u(t)}_{x_2(t)}$$

$$x_1(t) \longleftrightarrow \frac{-1}{S+2} \quad R_1 : \text{Re}\{S\} < -2$$

$$x_2(t) \longleftrightarrow \frac{1}{S-2} \quad R_2 : \text{Re}\{S\} > 2$$

$$\underbrace{x_1(t) + x_2(t)}_{B.S.S.} \longleftrightarrow R_1 \cap R_2 = \{\phi\} \quad X(S) \quad D.N.E.$$



Time shifting:

$$x(t) \longleftrightarrow X(S) \qquad ROC : R_1$$

$$x(t - t_0) \longleftrightarrow e^{-St_0}X(S) \qquad ROC : R_1$$

$$x(t + t_0) \longleftrightarrow e^{St_0}X(S) \qquad ROC : R_1$$

Note: "Above properties do not change pole diagram on S plane".

9. $x(t) = e^{-(t-5)}u(t-5)$

Find BLT?

Solution:

$$e^{-t}u(t) \longleftrightarrow \frac{1}{S+1} \qquad ROC : \text{Re}\{S\} > -1$$

$$e^{-(t-5)}u(t-5) \longleftrightarrow \frac{e^{-5S}}{S+1} \qquad ROC : \text{Re}\{S\} > -1$$

10. $x(t) = -e^{-(t+5)}u(-t-5) \xleftrightarrow{B.L.T.} ?$

Solution:

$$-e^{-t}u(-t) \longleftrightarrow \frac{1}{S+1} \qquad ROC : \text{Re}\{S\} < -1$$

$$e^{-(t+5)}u(-(t+5)) \longleftrightarrow \frac{e^{5S}}{S+1} \qquad ROC : \text{Re}\{S\} < -1$$

Multiplication with exponential:

$$x(t) \longleftrightarrow X(S) \qquad ROC : \text{Re}\{S\}$$

$$e^{S_0 t}x(t) \longleftrightarrow X(S - S_0) \qquad ROC : \text{Re}\{S - S_0\}$$

$$e^{-S_0 t}x(t) \longleftrightarrow X(S + S_0) \qquad ROC : \text{Re}\{S + S_0\}$$

Note: Pole diagram on S plane changes.

11. $e^{2t}u(t+5) \longleftrightarrow ?$

Solution:

$$u(t) \longleftrightarrow \frac{1}{S}$$

$$u(t+5) \longleftrightarrow \frac{e^{5S}}{S}$$

$$e^{2t}\underbrace{u(t+5)}_{RSS} \longleftrightarrow \frac{e^{5(S-2)}}{\underbrace{(S-2)}}$$

$$\mathrm{Re}\{S\} = 2$$

$$ROC: \mathrm{Re}\{S\} > 2$$

$$u(t) \longleftrightarrow \frac{1}{S} \qquad \mathrm{Re}\{S\} > 0$$

$$u(t+5) \longleftrightarrow \frac{e^{5S}}{S} \qquad \mathrm{Re}\{S\} > 0$$

$$e^{2t}u(t+5) \longleftrightarrow \frac{e^{5(s-2)}}{(S-2)} \qquad \mathrm{Re}\{S-2\} > 0$$

$$\mathrm{Re}\{S-2\} > 0$$

$$\mathrm{Re}\{\sigma + j\omega - 2\} > 0 \equiv \sigma - 2 > 0$$

$$\sigma > 2 : \mathrm{Re}\{S\} > 2$$

12. $x(t) = e^{2t}u(t) - e^{2t}u(t-5)$

Solution:

$$x(t) = \underbrace{e^{2t}u(t)}_{x_1(t)} - \underbrace{e^{2t}u(t-5)}_{x_2(t)}$$

$$e^{2t}u(t) \longleftrightarrow \frac{1}{S-2} \qquad R_1 : \mathrm{Re}\{S\} > 2$$

$$e^{2t}u(t-5) \longleftrightarrow \frac{e^{-5(S-2)}}{S-2} \qquad R_2 : \mathrm{Re}\{S\} > 2$$

$$\underbrace{x(t) = x_1(t) - x_2(t)}_{F.D. + Abs\int} \longleftrightarrow ROC: \text{entire } S \text{ plane}$$

$$X(S) = \left\{\frac{1 - e^{-5(S-2)}}{S-2}\right\}$$

$$x(t) = e^{2t}\left[u(t) - u(t-5)\right]$$

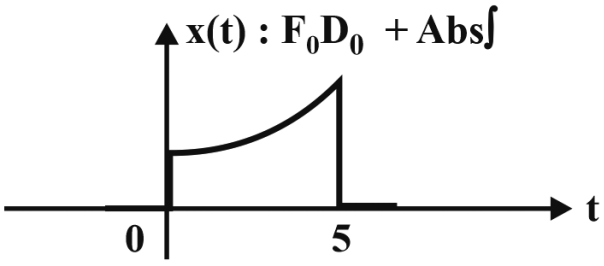


Fig. 9.15

Few important results:

$$x(t) \longleftrightarrow X(S) \qquad ROC: \mathrm{Re}\{S\}$$

$$e^{S_0 t}x(t) \longleftrightarrow X(S - S_0) \qquad ROC: \mathrm{Re}\{S - S_0\}$$

$$e^{-S_0 t}x(t) \longleftrightarrow X(S + S_0) \qquad ROC: \mathrm{Re}\{S + S_0\}$$

$$u(t) \longleftrightarrow \frac{1}{S} \qquad \mathrm{Re}\{S\} > 0$$

$$e^{-j\omega_0 t}u(t) \longleftrightarrow \frac{1}{S + j\omega_o} : \ \mathrm{Re}\{S + j\omega_0\}$$

$$\mathrm{Re}\{\sigma + j\omega + j\omega_0\} > 0$$

$$\sigma > 0$$

$$\mathrm{Re}\{S\} > 0 \rightarrow R_1$$

$$e^{j\omega_0 t}u(t) \longleftrightarrow \frac{1}{S - j\omega_o} : \ \mathrm{Re}\{S\} > 0 \rightarrow R_2$$

$$\cos\omega_0 t\, u(t) \longleftrightarrow \frac{S}{S^2 + \omega_0^2} \qquad ROC: \mathrm{Re}\{S\} > 0$$

$$\sin\omega_0 t\, u(t) \longleftrightarrow \frac{\omega_0}{S^2 + \omega_0^2} \qquad ROC: \mathrm{Re}\{S\} > 0$$

a : Real

$$e^{-at}\cos\omega_0 t\,u(t)\longleftrightarrow\frac{S+a}{(S+a)^2+\omega_0^2}\qquad ROC:\mathrm{Re}\{S\}>-a$$

$$e^{-at}\sin\omega_0 t\,u(t)\longleftrightarrow\frac{\omega_0}{(S+a)^2+\omega_0^2}\qquad ROC:\mathrm{Re}\{S\}>-a$$

a : real

$$1\xleftrightarrow{F.T.}2\pi\delta(\omega)$$

$$\cos\omega_0 t\xleftrightarrow{F.T.}\pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0)$$

$$\sin\omega_0 t\xleftrightarrow{F.T.}\frac{\pi}{j}\left(\delta(\omega-\omega_0)-\delta(\omega+\omega_0)\right)$$

$$\cos\omega_0 tu(t)\longleftrightarrow\frac{F(\omega-\omega_0)+F(\omega+\omega_0)}{2}$$

$$\sin\omega_0 tu(t)\longleftrightarrow\frac{F(\omega-\omega_0)+F(\omega+\omega_0)}{2j}$$

$$1\xleftrightarrow{BLT}Does\ Not\ Exists$$

$$\cos\omega_0 t\xleftrightarrow{BLT}Does\ Not\ Exists$$

$$\sin\omega_0 t\xleftrightarrow{BLT}Does\ Not\ Exists$$

$$\cos\omega_0 tu(t)\longleftrightarrow\frac{S}{S^2+\omega_0^2}\ :\ \sigma>0$$

$$\sin\omega_0 tu(t)\longleftrightarrow\frac{\omega_0}{S^2+\omega_0^2}\ :\ \sigma>0$$

$$u(t)\longleftrightarrow F(\omega)=\pi\delta(\omega)+\frac{1}{j\omega}$$

$$-u(-t)\longleftrightarrow\frac{1}{S}\ :\ \mathrm{Re}\{S\}<0$$

$$-e^{j\omega_0 t}u(-t)\longleftrightarrow\frac{1}{S-j\omega_0}\ :\ \mathrm{Re}\{S\}<0$$

$$-e^{-j\omega_0 t}u(-t)\longleftrightarrow\frac{1}{S+j\omega_0}\ :\ \mathrm{Re}\{S\}<0$$

$$-\cos\omega_0 tu(-t)\longleftrightarrow\frac{S}{S^2+\omega_0^2}\ :\ \mathrm{Re}\{S\}<0$$

$$-\sin\omega_0 tu(-t)\longleftrightarrow\frac{\omega_0}{S^2+\omega_0^2}\ :\ \mathrm{Re}\{S\}<0$$

$$-e^{-at}\cos\omega_0 tu(-t)\longleftrightarrow\frac{(S+a)}{(S+a)^2+\omega_0^2}\ :$$

$\mathrm{Re}\{S\}<-a$

$$-e^{-at}\sin\omega_0 tu(-t)\longleftrightarrow\frac{\omega_0}{(S+a)^2+\omega_0^2}\ :$$

$\mathrm{Re}\{S\}<-a$

Time reversal:

$$x(t)\longleftrightarrow X(S)\qquad ROC:\mathrm{Re}\{S\}$$

$$x(-t)\longleftrightarrow X(-S)\qquad ROC:\mathrm{Re}\{-S\}$$

Note: Pole diagram changes.

13. $u(-t)\longleftrightarrow$?

Solution:

$$u(t)\longleftrightarrow\frac{1}{S}\qquad \mathrm{Re}\{S\}>0$$

$$u(-t)\longleftrightarrow\frac{1}{(-S)}\qquad \mathrm{Re}\{-S\}>0$$

$$\mathrm{Re}\{-\sigma-j\omega\}>0$$

$$-\sigma>0$$

$$\sigma<0$$

$$\mathrm{Re}\{S\}<0$$

14. $x(t) = e^{-St}u(-t-5) \xleftrightarrow{BLT}$

Solution:

$$u(t) \xleftrightarrow{} \frac{1}{S} : \text{Re}\{S\} > 0$$

$$u(t-5) \xleftrightarrow{} \frac{e^{-5S}}{S} : \text{Re}\{S\} > 0$$

$$u(-t-5) \xleftrightarrow{} \frac{e^{5S}}{-S} : \text{Re}\{S\} > 0$$

$\neq 0$

$-\infty < t < -5$

L.S.S.

$$e^{-5t}u(-t-5) \xleftrightarrow{} \frac{e^{5(S+5)}}{-(S+5)}$$

$\text{Re}\{S+5\} < 0 \rightarrow \text{Re}\{S\} < -5$

t = −5

t = −ve

pole: S = −5

$Re\{s\} = -5 \rightarrow \text{Re}\{s\} < -5$

Time scaling:

$$x(t) \xleftrightarrow{} X(S) \qquad ROC : \text{Re}\{S\}$$

$$x(at) \xleftrightarrow{} \frac{1}{|a|}X\left(\frac{S}{a}\right) \qquad ROC : \text{Re}\left\{\frac{S}{a}\right\}$$

Note: Pole diagram changes.

15. $e^{t/3}u(t/3) \xleftrightarrow{BLT} ?$

Solution:

$$e^{t}u(t) \xleftrightarrow{} \frac{1}{S-1} \quad ROC : \text{Re}\{S\} > 1$$

$$a = 1/3 \quad e^{t/3}u(t/3) \xleftrightarrow{} \frac{1}{a}\left\{\frac{1}{\frac{S}{a}-1}\right\}$$

$$ROC : \text{Re}\left\{\frac{S}{a}\right\} > 1$$

$$\left\{\frac{3}{3S-1}\right\} \qquad ROC : \text{Re}\{3S\} > 1$$

$\text{Re}\{S\} > 1/3$

Multiplication by T Property or Differentiation in S-Domain

$$x(t) \xleftrightarrow{} X(S) \qquad ROC : \text{Re}\{S\}$$

$$tx(t) \xleftrightarrow{} -\frac{dX(s)}{ds} \qquad ROC : \text{Re}\{S\}$$

Note: F.T.

$$tx(t) \xleftrightarrow{} +\frac{jdX(\omega)}{d\omega} \qquad Z.T.$$

$$nx(n) \xleftrightarrow{} -\frac{zdX(z)}{dz}$$

L.T.

$$tx(t) \xleftrightarrow{} -\frac{dX(S)}{dS} \qquad DTFT$$

$$nx(n) \xleftrightarrow{} +\frac{jdX(\Omega)}{d\Omega}$$

16. $x(t) \xleftrightarrow{} \frac{1}{S^2+S+1} : \text{Re}\{S\} > -1/2$

Solution:

$$y(t) = tx(t) \xleftrightarrow{} Y(S) = ?$$

$$Y(S) = -\frac{dX(S)}{dS} =$$

$$-\left\{\frac{(S^2+S+1)\times 0 - (2S+1)}{(S^2+S+1)^2}\right\}$$

$$Y(S) = \frac{2S+1}{(S^2+S+1)^2}$$

$ROC : \text{Re}\{S\} > -1/2$

$u(t) \longleftrightarrow \frac{1}{S}\text{Re}\{s\} > 0$	$-u(-t) \longleftrightarrow \frac{1}{S}\text{Re}\{s\} < 0$
$t^n u(t) \longleftrightarrow \frac{n!}{s^{n+1}}\text{Re}\{s\} > 0$	$-t^n u(-t) \longleftrightarrow \frac{n!}{s^{n+1}}\text{Re}\{s\} < 0$
$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a}\text{Re}\{s\} > -a$	$-e^{-at}u(-t) \longleftrightarrow \frac{1}{S+a}\text{Re}\{s\} < -a$
$t^n e^{-at}u(t) \longleftrightarrow \frac{n!}{(s+a)^{n+1}}\text{Re}\{s\} > -a$	$-t^n e^{-at}u(-t) \longleftrightarrow \frac{n!}{(s+a)^{n+1}}\text{Re}\{s\} < -a$

"BLT of singular function".

$$u(t) \longleftrightarrow \frac{1}{S} \quad :\text{Re}\{S\} > 0$$

$$r(t) = tu(t) \longleftrightarrow \frac{1}{S^2} \quad :\text{Re}\{S\} > 0$$

$$p(t) = \frac{t^2}{2}u(t) \longleftrightarrow \frac{1}{S^3} \quad :\text{Re}\{S\} > 0$$

BLT of causal periodic signal:

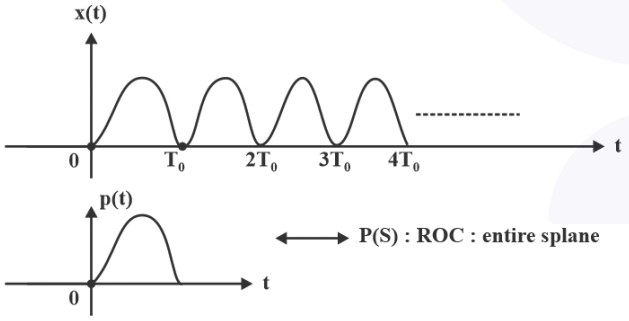


Fig. 9.16

S.2

$$x(t) = \underbrace{p(t) + p(t-T_0) + p(t-2T_0) + p(t-3T_0) + \cdots}_{RSS}$$

$$X(S) = P(S)\left[1 + e^{-ST_0} + e^{-S2T_0} + e^{-S3T_0} + \cdots\right]$$

$$X(S) = \frac{P(S)}{\left(1 - e^{-ST_0}\right)} \qquad ROC : Rightward.$$

Re{S}>0

Exact R.O.C:

$$\left|e^{-ST_0}\right| < 1 \qquad |z_1 z_2| = |z_1||z_2|$$

$$\left|e^{-\sigma T_0} e^{-j\omega T_0}\right| < 1$$

$$\left|e^{-T_0\sigma}\right| < 1$$

$e^{-T_0\sigma} < 1 \rightarrow$ only when $\sigma > 0$

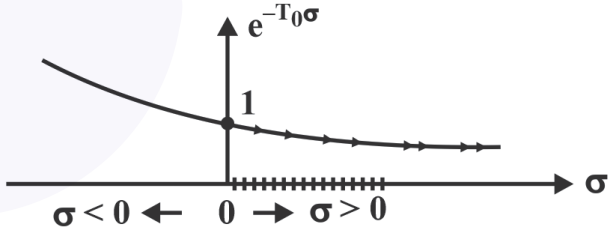


Fig. 9.17

17.

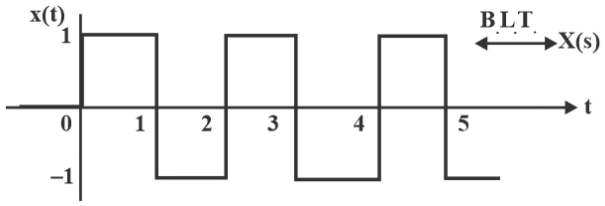


Fig. 9.18

Solution:

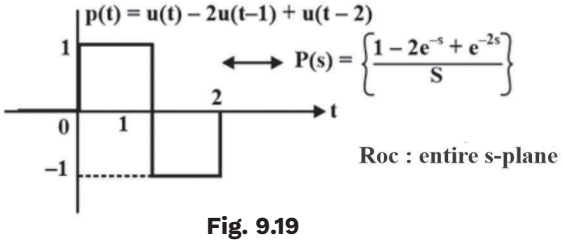


Fig. 9.19

$$x(t) = p(t) + p(t-2) + p(t-4) + \cdots$$

$$X(S) = P(S)\left[1 + e^{-2S} + e^{-4S} + \cdots\right]$$

$$X(S) = \frac{P(S)}{\left(1 - e^{-2S}\right)} \qquad ROC: \mathrm{Re}\{S\} > 0$$

ROC:

1. Each BLT has ROC associated with it
2. $x(t) \longleftrightarrow X(s)$ ROC: 2-D Region in the s plane

$$X(s)\Big|_{s=s_o} = \begin{cases} Finite\ s = s_o\ i\,s\ in\ ROC \\ Infinite\ s = s_o\ is\ not\ in\ ROC \end{cases}$$

3. BLT is finite, or BLT converges only for the range of σ values.
4. Each σ value represents a vertical line parallel to the jω axis.
5. Range of σ values consists of strip of parallel lines known as ROC

ROC building concept:

1. $x(t) = \delta(t) \longleftrightarrow X(s) = 1$

   → ROC of F.D. + Abs $\int$ Signal is always entire S plane

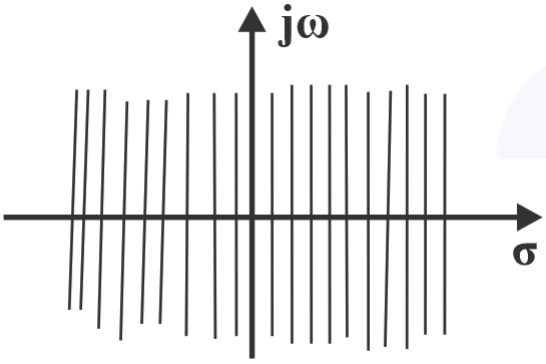


Fig. 9.20

If x(t) is absolutely integral, the ROC of X(s) includes the jω axis.

$x(t) = \delta(t) \longleftrightarrow x(s) = 1$ ROC: Entire S plane

$x(s = j\omega) = F.T\ OF\ x(t)$

Calculation of $F.T$ form $B.L.T$ is calculate by putting $S = j\omega$ is part of ROC.

2. $x(t) = e^{-2t}u(t)$

   a) RSS

   b) Absolutely integrable

$$x(t) = e^{-2t}u(t) \longleftrightarrow X(s) = \frac{1}{s+2}$$ ROC: Re {s} >-2

s=-2

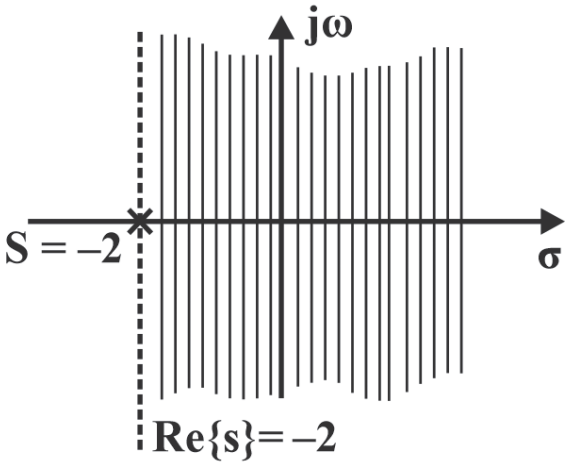


Fig. 9.21

Re(s) = –2

ROC : Re(s) > –2

Rightside of the right most pole (Rightward)

3. $x(t) = -e^{-2t}u(-t)$

   a) L.S.S

   b) Abs not integrable.

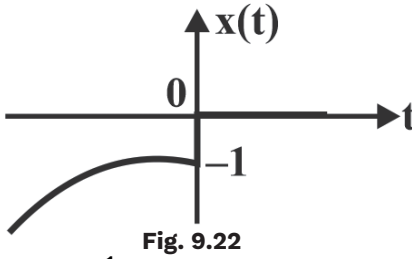


Fig. 9.22

$$x(t) \xleftarrow{BLT} \frac{1}{S+2} \quad ROC: \mathrm{Re}\{s\} < -2$$

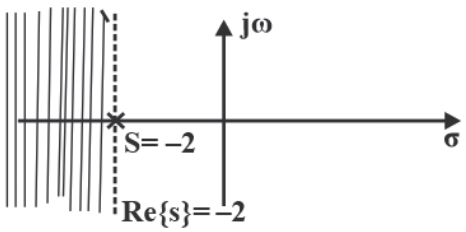


Fig. 9.23

Leftside of the left most pole (Leftward)

4. $x(t) = e^{-2|t|}$: Both sided

: Absolutely integrable

$$x(t) \longleftrightarrow X(s) = \frac{+1}{s+2} - \frac{1}{s-2}$$

ROC : $2 < \text{Re}\{s\} < 2$

strip form bounded by poles.

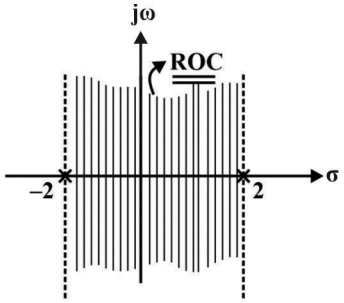


Fig. 9.24

Points to Remember

$$\underset{R.S.S.}{x(t) = e^{\frac{-t^2}{2}} u(t)} \xleftrightarrow{BLT} D.N.E$$

$$\underset{L.S.S.}{x(t) = -e^{\frac{-t^2}{2}} u(-t)} \xleftrightarrow{BLT} D.N.E$$

$$\underset{B.S.S.}{x(t) = 1} \xleftrightarrow{BLT} D.N.E$$

Properties of ROC

1. ROC may or may not include zeros of X(s)
2. ROC should never include poles of X(s)
3. ROC of finite duration and the absolutely integrable signal is the entire S plane
4. ROC of absolutely Integrable signal includes jω axis.
5. x(t): RSS
   - X(s): DNE
   - If X(s) converges, then ROC will be right-side of right most pole.
6. x(t): LSS
   - X(s) DNE
   - If X(s) exist, then ROC will be left side of leftmost pole.
7. x(t): BSS
   - X(s): DNE
   - If X(s) exist, ROC will be strip form bounded by poles.

Inverse Laplace Transform (ILT)

Building concepts:

18. $X(s) = \frac{1}{s+2}$ ROC: Re{s}>-2. Calculate ILT

Solution:

$$\underbrace{X(s) = \frac{1}{s+2}}_{Rational} : \underbrace{ROC : \text{Re}\{s\} > -2}_{Rightwards} \longleftrightarrow \underbrace{x(t) = e^{-2t}u(t)}_{R.S.S}$$

Causal

19. $X(s) = \frac{1}{S+2} : ROC : \text{Re}\{s\} < -2 \cdot$

Calculate ILT

Solution:

$$\underbrace{X(s) = \frac{1}{S+2}}_{Rational} : \underbrace{\text{Re}\{s\} < -2}_{Leftward} \longleftrightarrow \underbrace{x(t) = -e^{-2t}u(-t)}_{L.S.S}$$

Anticausal

20. $x(s) = \frac{e^s}{s+2}$ $ROC : \text{Re}\{s\} > -2$ ILT

Solution:

$$\underbrace{x(s) = \frac{e^s}{s+2}}_{Irrational} \underbrace{ROC : \text{Re}\{s\} > -2}_{Rightward} \longleftrightarrow$$

$$\underbrace{x(t) = e^{-2(t+1)}u(t+1)}_{R.S.S}$$

Noncausal

21. $X(s) = \frac{e^{-s}}{s+2}$ ROC: $\text{Re}\{s\} < -2$

Solution:

$$\underbrace{x(s) = \frac{e^{-s}}{s+2}}_{Irrational} \underset{Leftwards}{\text{Re}\{s\}} < -2 \longleftrightarrow x(t) =$$

$$\underset{L.S.S}{-e^{-2(t-1)}u(-t+1)}$$

Noncausal.

Points to Remember

X(s)	ROC	X(t)	Nature
$\frac{1}{s+2}$	Re{s} > –2	$e^{-2t}u(t)$	RSS
$\frac{1}{s+2}$	Re{s} < –2	$-e^{-2t}u(-t)$	LSS
$\frac{e^{s}}{s+2}$	Re{s} > –2	$e^{-2(t+1)}u(t+1)$	RSS
$\frac{e^{s}}{s+2}$	Re{s} < –2	$-e^{-2(t+1)}u(-t-1)$	LSS
$\frac{e^{-s}}{s+2}$	Re{s} > –2	$e^{-2(t-1)}u(t-1)$	RSS
$\frac{e^{-s}}{s+2}$	Re{s} < –2	$-e^{-2(t-1)}u(-t+1)$	LSS

Table 9.1

Conclusion: X(s): Rational–irrational

ROC	X(t)
Rightside of right most pole	RSS
Leftside of left most pole	LSS

Table 9.2

Conclusion: X(s): Rational

ROC	x(t)
Rightside of right most pole	Causal
Leftside of left most pole	Anticausal

Table 9.3

Steps to calculate ILT:

Given: Calculate ILT of $X(s)=\frac{N(s)}{D(s)}$

S.1 Common factor cancellation and pole calculation

S.2 Calculate the real part of poles → possible ROC

S.3 Calculate ILT corresponding to each possible ROC

22. $X(s)=\frac{1}{(S-1)(S+2)(S-4)^2}$:

Calculate x(t) =?

Solution:

S.1 $X(s)=\frac{1}{(S-1)(S+2)(S-4)^2}$

Poles: S=1 , S=-2, S=4(2 poles)

Re {s} = 1, Re {s} = - 2, Re {s} = 4

Possible ROCs:

a) Re {s} < - 2,

b) – 2 < Re {s} < 1

c) 1 < Re {s} < 4,

d) Re {s} > 4

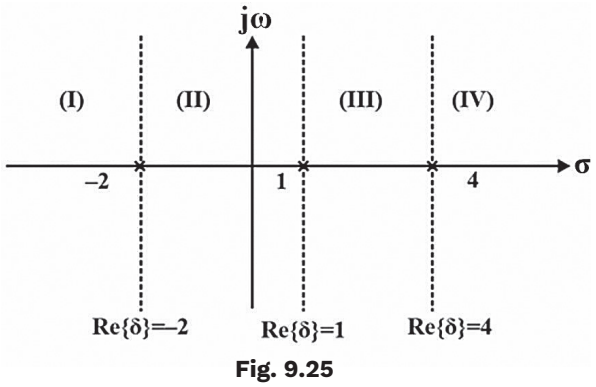


Fig. 9.25

a) R.O.C: Re {s} <-2 leftwards

$$X(s)=\frac{1}{(S-1)(S+2)(S-4)^2}=$$

$$\frac{a_0}{S+2}+\frac{a_1}{S-1}+\frac{a_2}{(S-4)}+\frac{a_3}{(S-4)^2}$$

Rational pole: S = - 2, S = 1, S = 4, S = 4

Re {s} = - 2 Re {s} = 1 Re {s} = 4 Re {s} = 4

Redefine ROC: Re {s} < - 2 Re {s} < 1 Re {s} < 4 Re {s} < 4

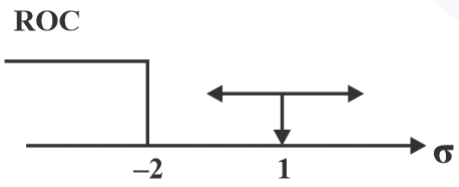


Fig. 9.26

$$x(t)=-a_0e^{-2t}u(-t)-a_1e^{t}u(-t)-a_2e^{4t}u(-t)-a_3te^{4t}u(-t)$$

Anticausal:

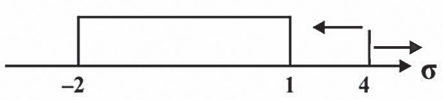


Fig. 9.27

b)

$-2 < \text{Re}\{s\} < 1$

strip

$$X(s)=\frac{a_0}{(S+2)}+\frac{a_1}{(S-1)}+\frac{a_2}{(S-4)}+\frac{a_3}{(S-4)^2}$$

Rational Re{s} = -2 Re{s} = 1 Re{s} = 4 Re{s} = 4

ROC: Re{s}>-2 Re{s} <1 Re{s} <4 Re{s} <4

$$x(t)=\underbrace{a_0e^{-2t}u(t)}-\underbrace{a_1e^{t}u(-t)-a_2e^{4t}u(-t)-a_3te^{4t}u(-t)}_{Noncausal}$$

c) 1 < Re {s} <4



Fig. 9.28

$$X(S)=\frac{a_0}{(S+2)}+\frac{a_1}{(S-1)}+\frac{a_2}{(S-4)}+\frac{a_3}{(S-4)^2}$$

Re{s} = -2 Re{S} = 1 Re{S} = 4 Re{S} = 4

RE{S} >- 2 Re{S} >1 Re{S} < 4 Re{S} < 4

$$x(t)=a_0e^{-2t}u(t)+a_1e^{t}u(-t)-a_2e^{4t}u(-t)-a_3te^{4t}u(-t)$$

d) Re {s}>4 Rightwards

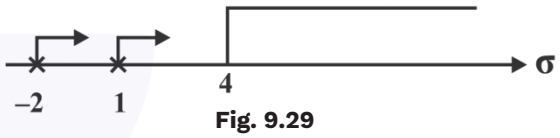


Fig. 9.29

$$X(S)=\frac{a_0}{(S+2)}+\frac{a_1}{(S-1)}+\frac{a_2}{(S-4)}+\frac{a_3}{(S-4)^2}$$

Rational Re{s} = -2 Re{s} = 1 Re{s} = 4 Re{s} = 4

ROC: RE{S} >- 2 Re{S} > 1 Re{S} > 4 Re{S} > 4

$$x(t)=a_0e^{-2t}u(t)+a_1e^{t}u(t)+a_2e^{4t}u(t)+a_3te^{4t}u(t)$$

23. $X(S)=\dfrac{1}{(S+2)(S-1)(S-4)^2}$

a) Calculate x(t) such that it is absolutely integrable.

$x(t):Abs\int\rightarrow ROC\ include\ j\omega\ axis$

R.O.C: 2 < Re{S} < 1

b) Calculate ILT such that ILT exist from $-\infty<t<\infty$ but it is not absolutely integrable.

ILT → x (t)

$x(t):Both\ sided\longleftrightarrow ROC:strip$

$x(t)$: *Not Abs Integrable* $\longleftrightarrow$

ROC : $j\omega$ *axis not include*

R.O.C: 1 < Re{S} < 4

c) Calculate ILT such that ILT is causal.

$x(t)$: *Causal* $\longleftrightarrow$ *i.* $X(S)$: *Rational*

ROC : *ii. Right of Rightmost pole.*

ROC: Re{S} > 4

d) Calculate ILT such that ILT is non-causal.

ILT non – causal $\longleftrightarrow$ $X(S)$: *Rational*

ROC- Strip

-2< Re{s}< 1 and 1< Re{s}< 4

e) Calculate ILT such that ILT is anti-causal.

ILT is anti-causal: ↔ X(S): Rational

ROC: Leftsideof Leftmost pole

ROC: Re{s}< -2

f) Calculate ILT such that ILT is having both LS and RS nature and $x(t)e^{-\sigma t}$ is absolutely integrable but x(t) is not absolutely integrable.

$x(t)$: *L.S.* + *R.S.* = *B.S.S.* $\longleftrightarrow$

ROC : *strip*(2 *possible ROC*)

$x(t)e^{-\sigma t}$ Absolutely integrable → X(S) should be finite (all possible ROC)

x(t) not absolutely integrable → ROC should not include jω axis.

ROC: 1 < Re{S} < 4

g) Calculate x(t) such that its FT exist for all values of ω.

→ Well defined F.T. exist.

→x(t) is abs integrable: ROC: include jω axis

ROC: -2 < Re{S} < 1

h) Calculate ILT such that ILT is finite duration and absolutely integrable.

F.D. + *Abs* $\int$ $\longrightarrow$ *ROC* : *Entire S plane*

NO R.O.C

Number of ROCs or no. of inverse Laplace transforms (N):

$$N = \left\{\frac{\text{No of distirct complex poles}}{2}\right\} + \begin{Bmatrix}\text{No. of distind} \\ \text{Real Poles}\end{Bmatrix} + 1$$

24.

$$X(S) = \frac{(S+1)}{(S^2+S+1)^2(S^3+1)(S^2+2S+1)(S^2+3S+2)}$$

No. of inverse Laplace transforms?

Solution:

$$X(S) = \frac{(S+1)}{(S^2+S+1)^2(S+1)(S^2-S+1)}$$

$$(S+1)^2(S+1)(S+2)$$

$$X(S) = \frac{(S+1)}{(S^2+S+1)^2(S^2-S+1)(S+1)^3(S+2)}$$

$$S = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}, -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$$

$$S = \frac{1}{2} \pm j\frac{\sqrt{3}}{2},$$

$$S = -1, -1, -1$$

$$S = -2$$

$$N = \frac{2}{2} + \frac{2}{2} + 1 + 1 + 1 = 5$$

Convolution property:

$x_1(t) \longleftrightarrow X_1(S)$ $\quad ROC : R_1$

$x_2(t) \longleftrightarrow X_2(S)$ $\quad ROC : R_2$

$x(t) = x_1(t) * x_2(t) \longleftrightarrow R_1 \cap R_2 =$ $\{\phi\} : X(S) : D.N.E.$

$R_1 \cap R_2 \neq \{\phi\} : X(S) = X_1(S) X_2(S)$

25. $y(t) = e^{-|t|/2} * e^{-t} u(t) = ?$

Solution:

Remember:

Method 1: Time-domain convolution.

Method 2:

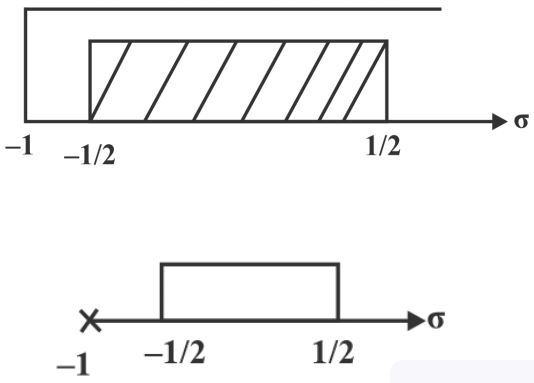


Fig. 9.30

ROC:

Differentiation property:

BLT ULT

Integration in time domain:

Conjugate:

Analysis of LTI system:

Stability of LTI System

1. For an LTI system to be stable, its unit impulse response h(t) must be absolutely Integration.
2. For h(t) to be absolutely Integration, the ROC of H(S) must include the jω axis.
3. For an LTI system to be stable, the ROC of H(S) must include the jω axis.

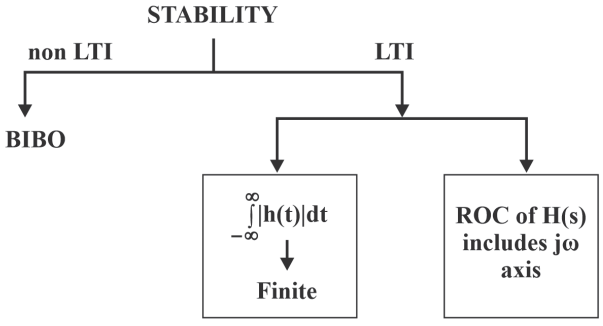


Fig. 9.31

Causality of an LTI system:

1. For an LTI system to be causal, h(t) must be a causal signal.
2. For a signal h(t) with rational H(S) to be causal, the ROC of H(S) should be right-side of the rightmost pole.
3. For an LTI system with Rational H(S), to be causal, the ROC of H(S) must be right-side of the rightmost pole.

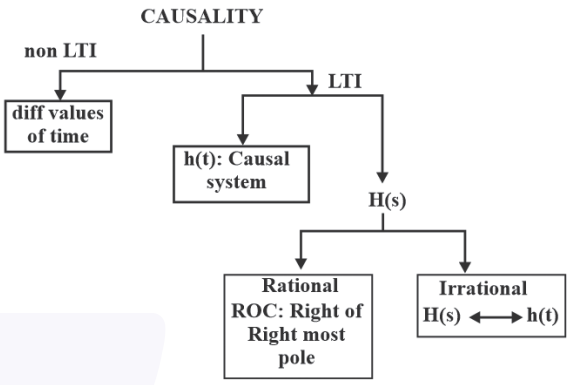


Fig. 9.32

26. The system has?

a) Stability

b) Causality

Solution:

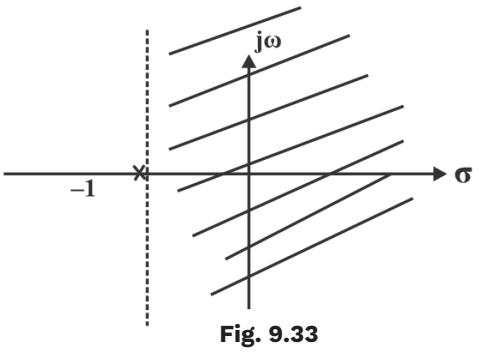


Fig. 9.33

S: LTI S/S is stable

C: LTI S/S is causal

Simultaneous Causal and Stable

1. For an LTI system to be causal, h(t) must be a causal signal. $h(t) = 0 \quad for \quad t < 0$
2. Rational H(S): all the poles of H(S) should be strictly in LHP3. Irrational H(s)

   Stability H(s) → ROC Check

   Causality H(s) ← h(t) : Check

27.

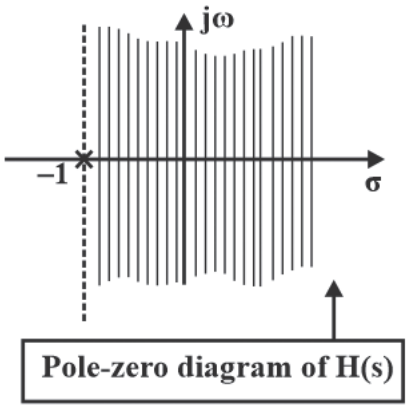


Fig. 9.34

System is possibly:

1. Causal and stable
2. Non-causal and stable
3. Anti-causal and stable
4. RSS and stable

Solution:

$$\text{Re}\{S\} > -1$$

$$H(S) = \frac{1}{S+1} \xleftrightarrow{\sigma > -1} h(t) = e^{-t}u(t)$$

$$H(S) = \frac{e^{S}}{S+1} \xleftrightarrow{\sigma > -1} h(t) = e^{-(t+1)}u(t+1)$$

$$H(S) = \frac{e^{-S}}{S+1} \xleftrightarrow{\sigma > -1} h(t) = e^{-(t+1)}u(t-1)$$

Points To Remember

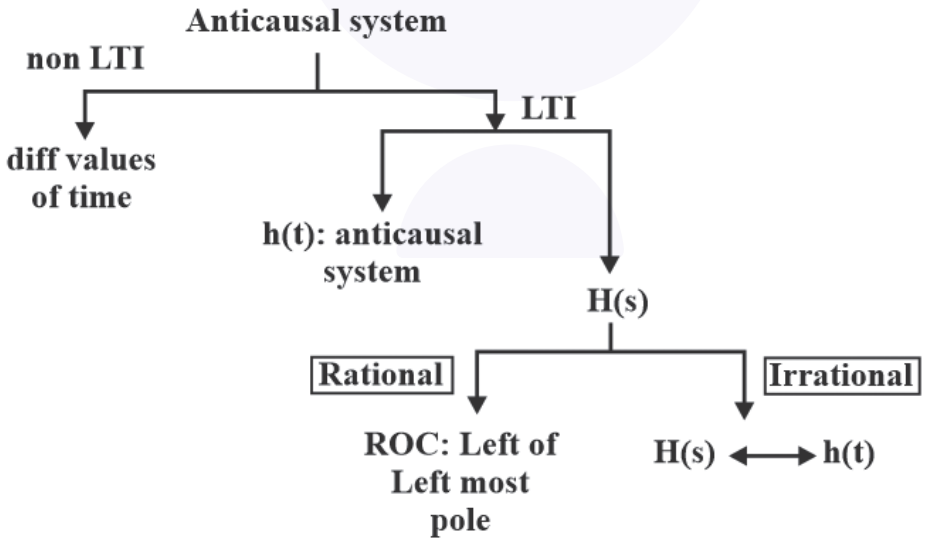


Fig. 9.35

Simultaneous Stable and Anti-Causal LTI S/S

1. $\left(h(t) = 0 : t \geq 0\right) \cap \left(\int_{-\infty}^{\infty} |h(t)|\,dt : \text{finite}\right)$
2. H(S) : Rational: All the poles of H(S) should be strictly on RHP3. H(s) Irrational
   Stability: ROC of H(s)
   Anticausality H(s) → h(t) → check h(t)

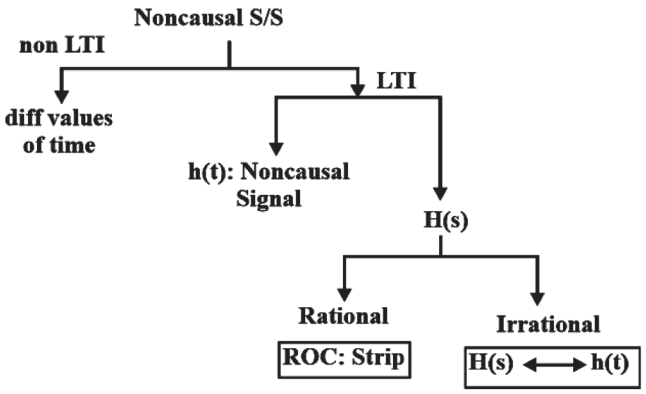


Fig. 9.36

Simultaneous stable and non-causal system:

1. $\left(h(t)=\text{non}-\text{causal}\right) \cap \left(\int_{-\infty}^{\infty}|h(t)|dt:\text{finite}\right)$
2. H(S) : Rational → Poles should be present on either side of jω axis.
3. H(s) : Irrational
   Stability : ROC of H(s)
   Non causality → H(s) ← h(t) : Check h(t)

Unilateral LT

$$X(S)=\int_{0^-}^{\infty}x(t)e^{-St}at$$

$$ULT\{x(t)\}=BLT\{x(t)u(t)\}=?$$

$$1\xleftrightarrow{U.L.T.}\frac{1}{S}$$

$$e^{-2t}\xleftrightarrow{U.L.T.}\frac{1}{S+2}$$

$$e^{2t}\xleftrightarrow{U.L.T.}\frac{1}{S-2}$$

$$\sin\omega_0 t\xleftrightarrow{U.L.T.}\frac{\omega_0}{S^2+\omega_0^2}$$

ROCs are not attached to ULT. Since, it is always rightwards.

$$\cos\omega_0 t\xleftrightarrow{U.L.T.}\frac{S}{S^2+\omega_0^2}$$

28. $e^{-2t}u(t+1)$

a) BLT

b) ULT

Solution:

a) $u(t)\longleftrightarrow\frac{1}{S}$

$$u(t+1)\longleftrightarrow\frac{e^S}{S}$$

$$e^{-2t}u(t+1)\longleftrightarrow\frac{e^{S+2}}{S+2}\qquad ROC:Re\{S\}>-2$$

b) $x(t)=e^{-2t}u(t+1)$

$$ULT\left[e^{-2t}u(t+1)\right]=BLT\left[e^{-2t}\underbrace{u(t+1)}_{-1\le t<\infty}\underbrace{u(t)}_{0\le t<\infty}\right]$$

$$=BLT\left[e^{-2t}u(t)\right]$$

$$=\frac{1}{S+2}$$

Note:

$x(t):\text{causal}:\quad BLT(x(t))=ULT(x(t))$

$x(t):\text{non}-\text{causal}:\quad BLT(x(t))\neq ULT(x(t))$

Properties of ULT:

Differentiation property:

$$\frac{dx(0^-)}{dt}=\left(\frac{dx(t)}{dt}\right)_{t=0^-}$$

$$x(t)\longleftrightarrow X(S)$$

$$\frac{dx(t)}{dt} \longleftrightarrow SX(S) - x(0^-)$$

$$\frac{d^2x(t)}{dt^2} \longleftrightarrow S\left[SX(S) - x(0^-)\right] - \frac{dx(0^-)}{dt}$$

$$S^2X(S) - Sx(0^-) - \frac{dx(0^-)}{dt}$$

$$\frac{d^3x(t)}{dt^3} \longleftrightarrow S^3X(S) - S^2x(0^-)$$
$$-S\frac{dx(0^-)}{dt} - \frac{d^2x(0^-)}{dt^2}$$

$$\boxed{\begin{matrix} x(0^-) \\ \frac{dx(0^-)}{dt} \\ \frac{d^2x(0^-)}{dt^2} \end{matrix}} \rightarrow \text{Initial values or Initial condition.}$$

Linear Constant Coefficient DE (LCCDE)

1. A differential equation represents a linear system. When
   a) Product terms of input, output and their derivatives are absent →

$$x(t)\cdot x(t),\ x(t)y(t),\ x(t)\frac{dy(t)}{dt},\ y(t)\frac{dy(t)}{dt}$$

   b) No constant addition.
2. A Differential Eqⁿ represent a time-invariant system if it is the constant coefficient. PE→

Example:

a) $\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + x(t)y(t)$
$= x(t)\begin{cases}\text{Non-linear}\\ \text{Time Invariant}\end{cases}$

b) $\frac{d^2y(t)}{dt^2} + \sin t\frac{dy(t)}{dt} + y(t)$
$= x(t)\begin{cases}\text{Linear}\\ \text{Time Variant}\end{cases}$

Case 1: D. Eqⁿ to transfer function:

Let an electrical system is represented by

$$\frac{d^2y(t)}{dt^2} + \frac{5dy(t)}{dt} + 6y(t) = \frac{2dx(t)}{dt} + 2x(t)$$

ULT on both sides

$$\frac{d^2y(t)}{dt^2} \longleftrightarrow S^2Y(S) - Sy(0^-) - \frac{dy(0^-)}{dt}$$

$$\frac{dy(t)}{dt^2} \longleftrightarrow SY(S) - y(0^-)$$

$$S^2Y(S) - Sy(0^-) - \frac{dy(0^-)}{dt} + 5SY(S)$$
$$-5y(0^-) + 6Y(S) = 2SX(S) - 2x(0^-) + 2X(S)$$

$$Y(S)\left[S^2 + 5S + 6\right] = \begin{Bmatrix} Sy(0^-) + \frac{dy(0^-)}{dt} \\ +5y(0^-) - 2x(0^-) \end{Bmatrix}$$
$$+X(S)(2S+2)$$

$$Y(S) = \frac{\left[Sy(0^-) + \frac{dy(0^-)}{dt} + 5y(0^-) - 2x(0^-)\right]}{S^2 + 5S + 6}$$
$$+ \underbrace{\frac{X(S)(2S+2)}{S^2 + 5S + 6}}_{ZSR}$$

ZIR

Transfer function:

$$\frac{Y(S)}{X(S)}\Big|_{IC=0} = H(S)$$

$$\frac{Y(S)}{X(S)} = \frac{2S+2}{S^2 + 5S + 6}$$

Case 2: TF to DE:

$$H(S) = \frac{2S+2}{S^2 + 5S + 6} = \frac{Y(S)}{X(S)}$$

$$S^2Y(S) + 5SY(S) + 6Y(S) = 2SY(S) + 2X(S)$$

$$\frac{d^2y(t)}{dt^2} + \frac{5dy(t)}{dt} + 6y(t) = \frac{2dx(t)}{dt} + 2x(t)$$

Note:

1. If the initial condition is zero:
   a) Transfer function can be calculated
   b) y(t) can be calculated from TFi.

$$\frac{Y(S)}{X(S)} = H(S)$$

   ii. $Y(S) = X(S)H(S)$
   iii. $y(t) \longleftrightarrow Y(S)$

2. Initial conditions are non-zero:
   a) T.F. can be calculated.
   b) y(t) cannot be calculated from TFi.

$$\frac{Y(S)}{X(S)} = H(S)$$

   ii. Form DE
   iii. ULT in D.E. and include I.C.

29. LTI S/S has $H(S) = \frac{S-1}{S+3}$ Initial condition $y(0^-) = 1$, $x(0^-) = 0$. Calculate y(t), when x(t)=u(t).

Solution:

Method 1: Wrong $\frac{Y(S)}{X(S)} = \frac{S-1}{S+3}$

$$Y(S) = \frac{S-1}{S+3}X(S) = \frac{S-1}{S(S+3)} = \frac{a_0}{S} + \frac{a_1}{S+3}$$

$$y(t) = (a_0 + a_1 e^{-3t})u(t)$$

Method 2: $\rightarrow \frac{Y(S)}{X(S)} = \frac{S-1}{S+3}$

$$SY(S) + 3Y(S) = SX(S) - X(S)$$

$$\rightarrow \frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} - x(t)$$

$$\rightarrow SY(S) - y(0^-) + 3Y(S) = SX(S) - x(0^-) - X(S)$$

$$SY(S) - 1 + 3Y(S) = SX(S) - X(S)$$

$$Y(S) = \frac{1}{S+3} + \frac{X(S)(S-1)}{S+3}$$

$$\boxed{Y(S) = \frac{1}{S+3} + \frac{(S-1)}{S(S+3)} = \frac{2S-1}{S(S+3)}} = \frac{a_0}{S} + \frac{a_1}{S+3}$$

$$Y(S) = -\frac{1}{3S} + \frac{7}{3(S+3)}$$

Taking ILT on both sides

$$y(t) = -\frac{1}{3}u(t) + \frac{7}{3}e^{-3t}u(t) \text{ SS}$$

Type of responses

Case 1: $x(t) = 0 \longrightarrow \underbrace{H(S)}_{IC \neq 0} \longrightarrow y(t)$

$= y_{ZIR}(t) =$ zero input response

$x(t) \neq 0 \longrightarrow \underbrace{H(S)}_{IC=0} \longrightarrow y(t)$

$= y_{ZSR}(t) =$ zero state response

$$\boxed{y_{total} = y_{ZIR}(t) + y_{ZSR}(t)}$$

Case 2:

$x(t) \rightarrow h(t) \rightarrow y_1(t)$: only input pole contribution

↓

Forced response.

$x(t) \rightarrow h(t) \rightarrow y_2(t)$: only system pole contribution

↓

Natural response.

30. $\frac{d^2y(t)}{dt^2} + \frac{5dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + 6x(t)$

Solution:

a) ZSR, ZIR

b) Forced, natural

$$\boxed{\begin{matrix} x(t) = u(t) \\ y(0^-) = 1 \\ \dot{y}(0^-) = 2 \end{matrix}} \longleftrightarrow X(S) = \frac{1}{S} \quad \text{Input pole : } S = 0$$

Initial condition→ Indication of ULT

$$S^2Y(S) - Sy(0^-) - \frac{dy(0^-)}{dt} + 5Sy(S)$$

$$-5y(0^-) + 6Y(S) = SY(S) - x(0^-) + 6X(S)$$

$$Y(S)\left[S^2+5S+6\right]=(S+7)+(S)(S+6)$$

$$Y(S)=\underbrace{\left\{\frac{S+7}{S^2+5S+6}\right\}}_{ZIR}+\underbrace{X(S)\left\{\frac{S+6}{S^2+5S+6}\right\}}_{ZSR}$$

$$Y(S)=\frac{S+7}{S^2+5S+6}+\frac{S+6}{S\left(S^2+5S+6\right)}$$

$$Y(S)=\left\{\frac{5}{S+2}-\frac{4}{S+3}\right\}+\left\{\frac{1}{S}-\frac{2}{S+2}+\frac{1}{S+3}\right\}$$

$$\downarrow$$

$$y(t)=\left\{5e^{-2t}u(t)-4e^{-3t}u(t)\right\}+\begin{Bmatrix}u(t)-2e^{-2t}u(t)\\+e^{-3t}u(t)\end{Bmatrix}$$

$\rightarrow$ Wrong

taking ILT on both sides

$$y(t)=\left\{5e^{-2t}-4e^{-3t}\right\}+\left\{1-2e^{-2t}+e^{-3t}\right\} \text{ for } t\geq 0^-$$

$$y(t)=\underbrace{\left\{5e^{-2t}-4e^{-3t}\right\}}_{ZIR}+\underbrace{\left\{1-2e^{-2t}+e^{-3t}\right\}}_{ZSR}$$

Taking LT on both sides

$$Y(S)=\left\{\frac{5}{S+2}-\frac{4}{S+3}\right\}+\left\{\frac{1}{S}-\frac{2}{S+2}+\frac{1}{S+3}\right\}$$

$$Y(S)=\underbrace{\frac{1}{S}}_{\substack{\text{Input}\\\text{Pole}}}+\underbrace{\left\{\frac{3}{S+2}-\frac{3}{S+3}\right\}}_{\text{System pole}}$$

Taking ILT on both sides

$$y(t)=\underbrace{1}_{\substack{\text{Forced}\\\text{response}}}+\underbrace{\left\{3e^{-2t}-3e^{-3t}\right\}}_{\text{Natural response}} \quad \text{for } t\geq 0^-$$

Initial Value Theorem of ULT

It is applicable when:

1. x(t) is causal.
2. x(t) does not have impulses or derivative of impulses at t = 0

And

$$X(S)=\frac{N(S)}{D(S)} \qquad \boxed{D(S)>N(S)} \; ***$$

$$\boxed{\lim_{t\to 0^+}x(t)=\lim_{s\to\infty}SX(S)}$$

$\rightarrow$ after pole zero canculation in $X(S)$

31. $X(S)=\frac{1}{S-1}$ Calculate the initial value of x(t)= ?

Solution:

$$\lim_{t\to 0^+}x(t)=\lim_{s\to\infty}SX(S)\rightarrow D^r>N^r$$

$$=\lim_{s\to\infty}\frac{S}{S-1}=\lim_{s\to\infty}\frac{S}{S\left(1-\frac{1}{S}\right)}=1$$

32. $X(S)=\frac{1}{S^2+4}$ Calculate IV of x(t) = ?

Solution:

$$X(S)=\frac{1}{S^2+4}$$

$$\downarrow$$

$$D^r>N^r$$

$$\lim_{t\to 0^+}x(t)=\lim_{s\to\infty}SX(S)=\lim_{s\to\infty}\frac{S}{S^2+4}=0$$

Final Value Theorem of ULT

It is applicable only when

1. x(t) is causal.

$$\boxed{\lim_{t\to 0^+}x(t)=\lim_{S\to\infty}SX(S)}$$

$\rightarrow X(S)$: Pole zero calculation.

Case 1: All poles of SX(S) strictly in LHP.

1. FV is finite
2. FVT is valid.

Case 2: If the pole location of SX(S) is

1. FV is indeterminate
2. FV is invalid.

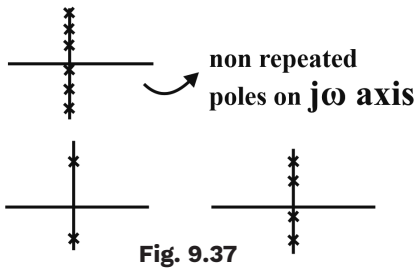


Fig. 9.37

Case 3: In other cases:

1. FV is ∞
2. FVT is not applicable.

33. Calculate the final value of x(t) if X(s) is given as follows.

a) $X(S)=\frac{1}{S+1}$

b) $X(S)=\frac{1}{S^2+1}$

Solution:

a) $X(S)=\frac{1}{S+1}$ $SX(S)\rightarrow$ all pole LHP

$x(\infty)=\lim_{S\to 0} SX(S)=0$

b) $X(S)=\frac{1}{S^2+1}$

$SX(S)\rightarrow$ not all pole in LHP

i. FV → Indeterminate

ii. FVT → X

$$\frac{1}{S^2+1}\begin{cases}\sin tu(t)\xrightarrow{t=\infty}-1\leq\ \leq 1\\ -\sin tu(-t)\xrightarrow{t=\infty}0\end{cases}$$

34. $X(S)=\frac{1}{S-1}$

Solution: $SX(S)\rightarrow$ not all pole in LHP

1. FV →∞
2. FVT → X

$$\frac{1}{S-1}\begin{cases}e^{t}u(t)\xrightarrow{t\to\infty}\infty\\ -e^{t}u(-t)\xrightarrow{t\to\infty}0\end{cases}$$

H(S): Rational/irrational:

ROC of $H(S)$	$h(t)$
Right of right most pole	R.S.S.
left of left most pole	L.S.S.
Strip	B.S.S.
Entire S plane	F.D. + Ab $\int$

H(S): Rational:

ROC of $H(S)$	$h(t)$
Right of right most pole	Causal
left of left most pole	Anti – causal
Strip	Non – causal
Entire S plane	F.D. + Ab $\int$

Block Diagram Representation

1. When a large number of components or elements are interconnected to form the complete system. It is convenient to represent the system by suitably interconnected subsystems, each of which can be separately and easily analysed.
2. A linear system can of a system with a transfer function H(s). The figure shows a block diagram respectively.

Cascade interconnection:

When the systems are connected in series, as depicted in fig. b), then the transfer function of the overall system is the product of the individual transfer functions. This result can also be proved by observing that in fig. b)

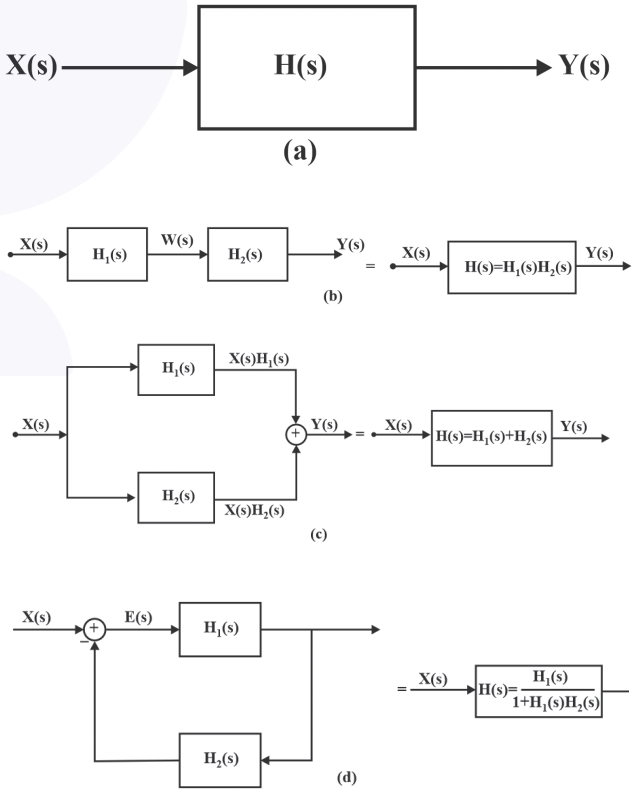


Fig. 9.38 Elementary Connection of Block Diagrams

Parallel Interconnection

When the systems are connected in parallel, as depicted in Fig. c), then the transfer function of the overall system is the sum of the individual transfer functions.

We can extend this result to any number of systems connected in parallel.

Feedback Interconnection

1. When the output is fed back to the input, as shown in Fig. d), the overall transfer function H(s) = Y(s)/X(s) can be computed as follows.
2. The input to the adder is X(s) and $-H_2(s)$ Y(s). Therefore, E(s), the output of the adder, is

   And the output is given by:

   (A)

   Therefore, the feedback loop can be replaced by a single block with the transfer function shown in Eq. (A). The transfer function $H_1(s)$ is the forward path transfer function, and $H_2(s)$ is the feedback path transfer function.

System Realisation

We develop a systematic method for the realisation (or implementation) of an arbitrary Nth-order transfer function. The most general transfer function with M=N is given by

1. There is no unique way of realising a system. A given transfer function can be realised in many different ways.

We avoid the use of differentiators because:

1. A differentiator represents and unstable system because a bounded input like the step input u(t) results in an unbounded output δ(t).

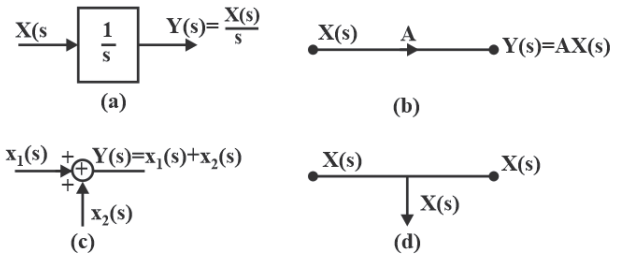


Fig. 9.39 Basic Building Block Representing a) Integrator, b) Multiplier, c) Adder, and d) Pick-Off Node

2. The noise is enhanced by a differentiator.

Direct form I realisation:

1. An Nth-order transfer function is characterised by 2N+1 unique coefficients and, in general, requires 2N+1 multipliers and 2N two-input adders for implementation.
2. Structures in which the multiplier coefficients are precisely the coefficients of the transfer function are called direct form structures.

Consider for simplicity a third-order system characterised by a transfer function.

We can express H(s) as

We can realise H(s) as a cascade of transfer function $H_1(s)$ followed by $H_2(s)$, as depicted in below fig. a), where

And

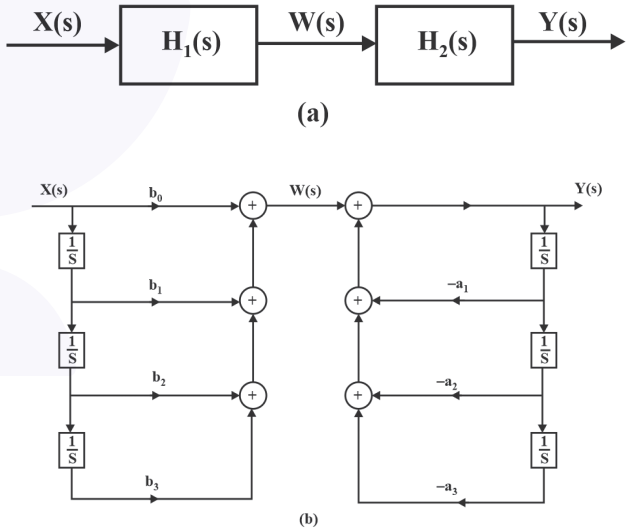


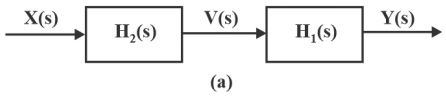
Fig. 9.40 a) Realisation of a Function in Two steps and b) Direct Form I Realisation of a Third-Order Continuous-Time LTI System

We can generalise this procedure, known as the direct form I realisation, for any value of N. This procedure requires 2N integrators to realise an Nth-order transfer function.

Direct form II (or canonic) realisation:

1. In the direct form I, we realise H(s) by implementing $H_1(s)$ followed by $H_2(s)$, as shown in above Fig. a).
2. We can also realise H(s), as shown in below Fig. a), where $H_2(s)$ is followed by $H_1(s)$.

3. This procedure is known as the direct form II realisation.



(a)

Fig. 9.41 a)

4. The direct form I realisation in above fig. b) implements zero first [the left-half section represented by $H_1(s)$] followed by realisation of poles [the right-half section represented by $H_2(s)$] of H(s).
5. In contrast, the direct form II realisation implements poles first followed by zeros.

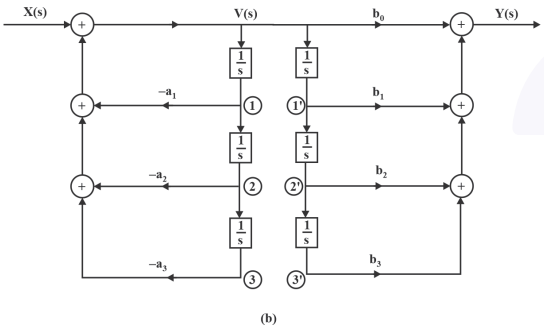
Again consider for simplicity a third-order system characterised by a transfer function.

We can express H(s) as

We can realise H(s) as a cascade of transfer function $H_2(s)$ followed by $H_1(s)$, as depicted in fig a), where

And

The left-half section of fig. b) shows the realisation of $H_2(s)$, and the right-half section shows the realisation of $H_1(s)$.



(b)

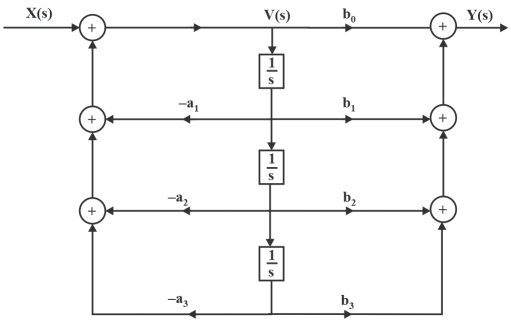


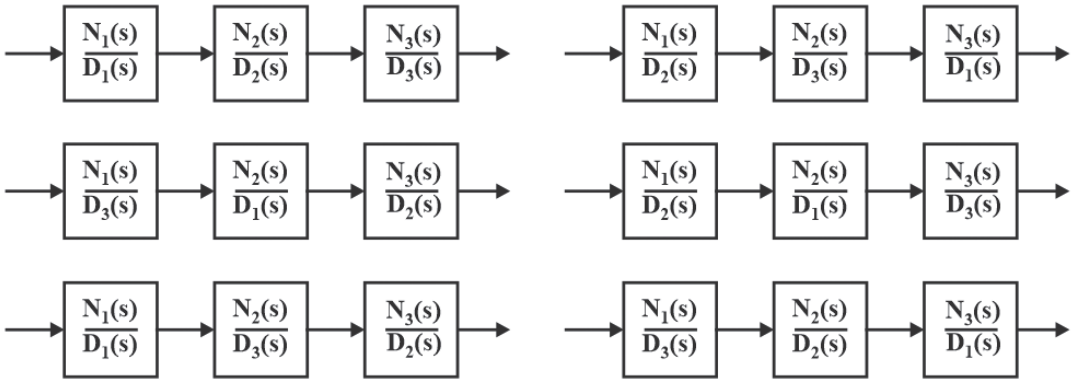
Fig. 9.42 Direct Form II Realisation of a Third-Order LTI System

This implementation halves the number of integrators to N and is thus more efficient in hardware utilisation. This is the direct form II realisation. This realisation is canonic since it employs N integrators to implement an Nth-order transfer function.

Cascade and parallel realisation:

1. By expressing the numerator polynomial N(s) and the denominator polynomial D(s) of the transfer function H(s) as a product of polynomials of lower degrees, a digital filter is often realised as a cascade of low-order systems.
2. Consider, for example, H(s) = N(s)/D(s) expressed as
3. Various different cascade realisations of H(s) can be obtained by different pole-zero polynomial pairings. Some examples of such realisations are shown in fig. a).

A system can be realised in a parallel form by making use of partial fraction expansion of the transfer function.



(a)

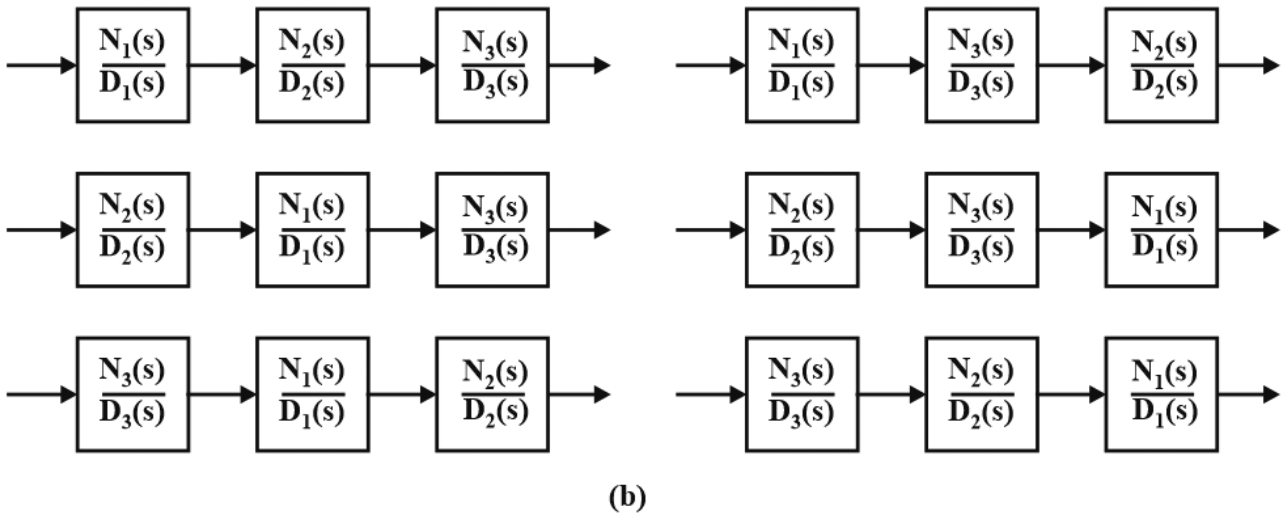


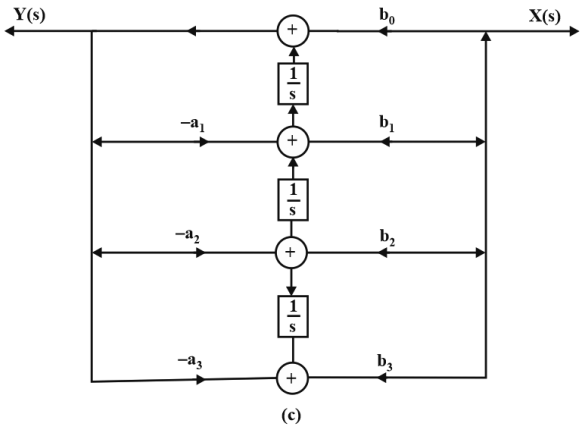
Fig. 9.43 Different Cascade Realisations Obtained by a) Different Pole-zero Pairings and b) Changing the Ordering of the Systems

Transposed Realisation

Two realisations are said to be equivalent if they have the same transfer function. A simple way to generate an equivalent structure from a given realisation is via the transpose operation, which is as follows:

1. Reverse all paths.
2. Replace pick-off nodes with adders and vice versa.
3. Interchange the input and output nodes.

Figure shows the transposed version of the canonic direct form realisation



Transposed version of the canonic direct from-II realization

Fig. 9.44

9.1 Find the canonic realisation of the following transfer functions:

1. $H(s)=\frac{s+5}{s+7}$
2. $H(s)=\frac{4s+28}{s^2+6s+5}$

Solution:

1. Given that

$$H(s)=\frac{Y(s)}{X(s)}=\frac{V(s)}{X(s)}\frac{Y(s)}{V(s)}$$

$$=\frac{s+5}{s+7}=\frac{1+\frac{5}{s}}{1+\frac{7}{s}}=\underbrace{\left(\frac{1}{1+\frac{7}{s}}\right)}_{H_2(s)}\underbrace{\left(1+\frac{5}{s}\right)}_{H_1(s)}$$

Where

$$H(s)==\frac{V(s)}{X(s)}=\frac{1}{1+\frac{7}{s}}$$

$$V(s)=X(s)-7\frac{V(s)}{s}$$

And

$$H_1(s)==\frac{Y(s)}{V(s)}=1+\frac{5}{s}$$

$$Y(s)=V(s)+5\frac{V(s)}{s}$$

The given transfer function H(s) = (s+5)/(s+7) is of the first order; therefore, we need only one integrator for its realisation, as shown in fig. a).

2. Given that

$$H(s)=\frac{Y(s)}{X(s)}=\frac{V(s)}{X(s)}\frac{Y(s)}{V(s)}=\frac{4s+28}{s^2+6s+5}$$

$$=\frac{\frac{4}{s}+\left(\frac{28}{s^2}\right)}{1+\left(\frac{6}{s}\right)+\frac{5}{s^2}}=\underbrace{\left(\frac{1}{1+\left(\frac{6}{s}\right)+\left(\frac{5}{s^2}\right)}\right)}_{H_2(s)}\underbrace{\left(\frac{4}{s}+\frac{28}{s^2}\right)}_{H_1(s)}$$

Where

$$H_2(s)=\frac{V(s)}{X(s)}=\frac{1}{1+\left(\frac{6}{s}\right)+\left(\frac{5}{s^2}\right)}$$

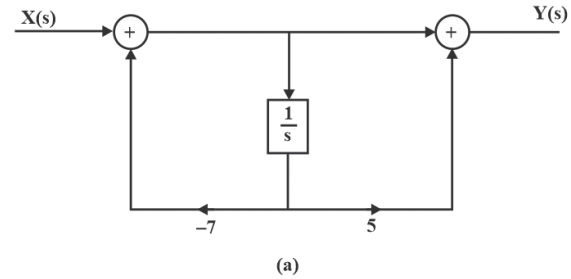
$$V(s)=X(s)-6\frac{V(s)}{s}-5\frac{V(s)}{s^2}$$

And

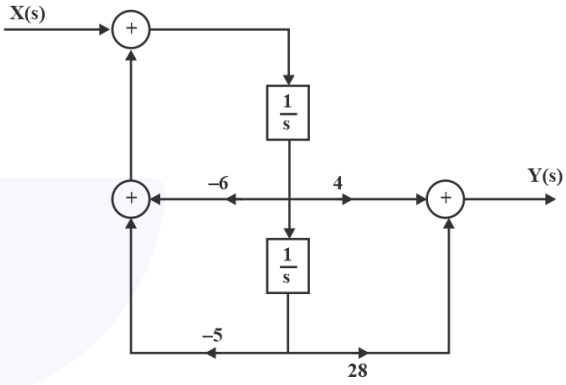
$$H_1(s)=\frac{Y(s)}{V(s)}=\frac{4}{s}+\frac{28}{s^2}$$

$$Y(s)=4\frac{V(s)}{s}+28\frac{V(s)}{s^2}$$

The given transfer function is of the second order; therefore, we need only two integrators for its realisation, as shown in Fig b).



(a)



(b)

Fig. 9.45

Exercise-Level 1

1. Let Y(s) be the unit step response of causal system having transfer function, $G(s)=\frac{3-s}{(s+1)(s+3)}$ that is $Y(s)=\frac{G(s)}{s}$. The forced response of system is:

(A) $u(t)-2e^{-t}u(t)+e^{-3t}u(t)$

(B) $2u(t)-2e^{-t}u(t)+e^{-3t}u(t)$

(C) $u(t)$

(D) $2u(t)$

2. The response of the system $G(s)=\frac{s-2}{(s+1)(s+3)}$ to the unit step input u(t) is y(t). The value of $\frac{dy}{dt}$ at $t=0^+$ is _____.

3. Let the signal $f(t)=0$ outside the interval $[T_1,T_2]$, were T_1 and T_2 are finite. Furthermore, $|f(t)|<\infty$. The region of convergence (R.O.C) of the signal's bilateral Laplace transform F(s) is

(A) A parallel strip containing the jΩ axis.

(B) A parallel strip not containing the jΩ axis.

(C) The entire s- plane.

(D) A half plane containing the jΩ axis.

4. The bilateral Laplace transform of a function $f(t)=\begin{cases}1, \text{if } a\le t\le b\\ 0, \text{otherwise}\end{cases}$ is

(A) $\frac{a-b}{s}$ (B) $\frac{e^z(a-b)}{s}$

(C) $\frac{e^{-as}-e^{-bs}}{s}$ (D) $\frac{e^{-(a-b)}}{s}$

5. Input x(t) and output y(t) of a LIT system are related by the differential equation $y''(t) - y'(t) - 6y(t) = x(t)$. If the system is neither causal nor stable, the impulse response h(t) of the system is

(A) $\frac{1}{5}e^{3t}u(-t) + \frac{1}{5}e^{-2t}u(-t)$

(B) $-\frac{1}{5}e^{3t}u(-t) + \frac{1}{5}e^{-2}u(-t)$

(C) $\frac{1}{5}e^{3t}u(-t) - \frac{1}{5}e^{-2t}u(t)$

(D) $-\frac{1}{5}e^{3t}u(-t) - \frac{1}{5}e^{-2t}u(t)$

Exercise-Level 2

1. Match List – I [Function in time domain f(t)] with List-II (Property) and select the correct answer using the code given below the lists:

List I

A. $\sin\omega_0 t u(t - t_0)$

B. $\sin\omega_0(t - t_0)u(t - t_0)$

C. $\sin\omega_0(t - t_0)u(t)$

D. $\sin\omega_0 t u(t)$

List II

1. $\frac{\omega_0}{s^2 + \omega_0^2}$

2. $\left\{\frac{\omega_0}{s^2 + \omega_0^2}\right\}e^{-t_0 s}$

3. $\frac{e^{-t_0 s}}{\sqrt{s^2 + \omega_0^2}}\sin\left(\omega_0 t_0 + \tan^{-1}\frac{\omega_0}{s}\right)$

4. $\frac{1}{\sqrt{s^2 + \omega_0^2}}\sin\left(\omega_0 t_0 - \tan^{-1}\frac{\omega_0}{s}\right)$

Codes:

	A	B	C	D
(A)	3	1	4	2
(B)	4	2	3	1
(C)	3	2	4	1
(D)	4	1	3	2

2. The transfer function of a causal LTI system is $H(s) = \frac{1}{s}S$. If the input to the system is $x(t) = \left[\frac{\sin t}{\pi t}\right]u(t)$, where u(t) is a unit step function, the system output y(t) as $t \to \infty$ _____.

3. The Laplace transform of the causal periodic square wave of period T shown in the figure below is

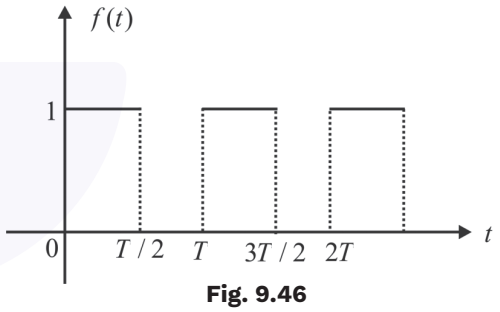


Fig. 9.46

(A) $F(s) = \frac{1}{1 + e^{-sT/2}}$

(B) $F(s) = \frac{1}{s(1 + e^{-sT/2})}$

(C) $F(s) = \frac{1}{s(1 - e^{-sT})}$

(D) $F(s) = \frac{1}{1 - e^{-sT}}$

4. The value of the integral

$\int_{-\infty}^{\infty} 12\cos(2\pi t)\frac{\sin(4\pi t)}{4\pi t}dt$ is _____.

5. Let $x(t) = \alpha s(t) + s(-t)$ with $s(t) = \beta e^{-4t}u(t)$, where u(t) is the unit step function . If the bilateral Laplace transform of x(t) is

$$X(s) = \frac{16}{s^2 - 16}, \quad -4 < \text{Re}\{s\} < 4;$$

Then the value of β is _____.

Answer Key

Level 1									
1.	(A)	2.	1	3.	(C)	4.	(C)	5.	(B)
Level 2									
1.	(C)	2.	0.5	3.	(B)	4.	3.	5.	2.

Solution Level – 1

1. (A)

Given: CLTF is given as, $G(s) = \dfrac{(3-s)}{(s+1)(s+3)}$

Output is given as $Y(s) = \dfrac{G(s)}{s}$

Let $Y(s) = \dfrac{(3-s)}{s(s+1)(s+3)}$

$$Y(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3}$$

$$A = \left.\frac{(3-s)}{(s+1)(s+3)}\right|_{S=0} = 1$$

$$B = \left.\frac{(3-s)}{s(s+3)}\right|_{S=-1} = -2$$

$$C = \left.\frac{(3-s)}{s(s+1)}\right|_{S=-3} = 1$$

$$Y(s) = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+3}$$

Taking inverse, Laplace transforms both sides.

$$y(t) = u(t) - 2e^{-t}u(t) + e^{-3t}u(t)$$

So, the forced response of the system is

$$y(t) = u(t) - 2e^{-t}u(t) + e^{-3t}u(t)$$

Given that the system is causal, so its impulse response will be zero for $t < 0$ i.e.

$$g(t) = 0, t < 0$$

As we have not considered any initial conditions, so the response $y(t)$ obtained is due to applied input only. Hence $y(t)$ itself represents the forced or zero state response.

Hence, the correct option is (A).

2. (1)

Given: $G(s) = \dfrac{s-2}{(s+1)(s+3)}$

Here, $U(s) = \dfrac{1}{s} = L[u(t)]$

$$G(s) = \frac{Y(s)}{U(s)}$$

$$Y(s) = G(s) \times \frac{1}{s} = \frac{(s-2)}{s(s+1)(s+3)}$$

$$Y(s) = \frac{(s-2)}{s(s+1)(s+3)}$$

$$Y(s) = \frac{-2}{3s} + \frac{3}{2(s+1)} - \frac{5}{6(s+3)}$$

Taking inverse Laplace transform,

$$y(t) = \frac{-2}{3} + \frac{3}{2}e^{-t} - \frac{5}{6}e^{-3t}$$

$$\frac{dy(t)}{dt} = \frac{-3}{2}e^{-t} + \frac{5}{2}e^{-3t}$$

$$\left.\frac{dy}{dt}\right|_{t=0} = \frac{-3}{2} + \frac{5}{2} = \frac{2}{2} = 1$$

Hence, the value of $\dfrac{dy}{dt}$ at $t = 0^+$ is 1.

Solution Level – 2

1. (C)

$$\sin \omega_0 t u(t) \rightarrow \frac{\omega_0}{s^2+\omega_0^2}$$

$$\sin_0(t-t_0)u(t-t_0) \rightarrow \left(\frac{\omega_0}{s^2+\omega_0^2}\right)e^{-t_0 s}$$

$$\sin \omega_0 t u(t-t_0) = \sin \omega_0 (t-t_0+t_0)u(t-t_0)$$

$$= \cos \omega_0 t_0 e^{-t_0 s}\frac{\omega_0}{s^2+\omega_0^2} + \sin \omega_0 e^{-t_0 s}\frac{s}{s^2+\omega^2}$$

$$= \cos \omega_0 t_0 e^{-t_0 s}\frac{\omega_0}{s^2+\omega_0^2} + \sin \omega_0 t_0 e^{-t_0 s}\frac{s}{s^2+\omega^2}$$

$$= \frac{1}{s^2\omega_0^2}e^{-t_0 s}\left\{s\cdot \sin \omega_0 t_0 + \omega_0 \cos \omega_0 t_0\right\}$$

Let $s = \cos\theta$

$\omega_0 = \sin\theta$

$$\theta = \tan^{-1}\left(\frac{\omega_0}{s}\right)$$

$$\sin \omega_0 t u(t-t_0) \rightarrow \frac{e^{-t_0 s}}{\sqrt{s^2+\omega_0^2}}$$

$$\sin\left[\omega_0 t_0 + \tan^{-1}\left(\frac{\omega_0}{s}\right)\right]$$

Similarly, we can find $\sin \omega_0(t-t_0)u(t)$

$$= \sin \omega_0 t \cos \omega_0 t_0 u(t) - \cos \omega_0 t \sin \omega_0 t_0 u(t)$$

$$= \frac{\omega_0}{\omega_0^2+s^2}\cos \omega_0 t_0 - \sin \omega_0 t_0 \frac{s}{s^2+\omega_0^2}$$

$$= -\frac{1}{\sqrt{s^2+\omega_0^2}}\sin\left(\omega_0 t_0 - \tan^{-1}\frac{\omega_0}{s}\right)$$

2. (0.5)

Given: $x(t) = \frac{\sin t}{\pi t}u(t), H(s) = \frac{1}{s}$

$$\sin t\, u(t) \xleftrightarrow{LT} \frac{1}{s^2+1}$$

From the property of Laplace transform,

$$f(t) \xleftrightarrow{LT} F(s)$$

$$\frac{f(t)}{t} \xleftrightarrow{LT} \int_s^\infty F(s)ds$$

Then, $\frac{\sin t}{t}u(t) \xleftrightarrow{LT} \int_s^\infty \frac{ds}{s^2+1}$

$$\frac{\sin t}{\pi t}u(t) \xleftrightarrow{LT} \frac{1}{\pi}\int_s^\infty \frac{ds}{s^2+1}$$

$$X(s) = \frac{1}{\pi}\int_s^\infty \frac{ds}{s^2+1} = \frac{1}{\pi}\left[\tan^{-1} s\right]_s^\infty$$

$$X(s) = \frac{1}{\pi}\left[\frac{\pi}{2} - \tan^{-1} s\right]$$

Output of given LTI system,

$y(t) = x \otimes h(t)$

Taking Laplace to transform,

$Y(s) = X(s)H(s)$

$$Y(s) = \frac{1}{s\pi}\left[\frac{\pi}{2} - \tan^{-1} s\right]$$

$$Y(s) = \frac{1}{2s} - \frac{\tan^{-1} s}{s\pi}$$

From the final value theorem,

$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$

$$y(\infty) = \lim_{s\to 0} s\left[\frac{1}{2s} - \frac{\tan^{-1} s}{s\pi}\right]$$

$$y(\infty) = \lim_{s\to 0} s\left[\frac{1}{2} - \frac{\tan^{-1} s}{\pi}\right]$$

$$y(\infty) = \frac{1}{2} - 0 = 0.5$$

Hence, the system output y(t) as $t \to \infty$ is 0.5

3. (B)

Laplace transform of a periodic signal is given by,

$$F(s) = L[f(t)] = \frac{F_1(s)}{1-e^{-sT}} = \frac{\int_0^{T/2} f_1(t)e^{-st}dt}{1-e^{-sT}}$$

Where, $F_1(s)$ is Laplace transform of the first period.

$$L[f(t)] = \frac{\int_0^{T/2} e^{-st}dt}{1-e^{-sT}} = \frac{1}{(1-e^{-sT})}\left[\frac{e^{-st}}{-s}\right]_0^{T/2}$$

$$L[f(t)] = \frac{1}{s(1-e^{-sT})}(1-e^{-sT/2})$$

$$L[f(t)] = \frac{1}{s}\frac{1-e^{-sT/2}}{s(1-e^{-sT/2})(1+e^{-sT/2})}$$

$$L[f(t)] = \frac{1}{s}\frac{1}{\left[1+e^{-sT/2}\right]}$$

Hence, the correct option is (B).